

Uniform Bessel bounds for endpoint crossings in the persistent random walk

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Abstract

A persistent random walk takes N steps of ± 1 , and each step after the first repeats the one before it with probability p . Its steps form the zero-field Ising chain with free boundaries. Kagey asked when its middle bin and its endpoints are equally likely. We compare the ratio of each bin to an endpoint with an explicit Bessel function, with relative error at most $Nu^2/2 + u$ for every bin and every N , where $u = (1 - p)/p$. At the center this brackets the crossing p_N for every $N \geq 2$. Put $L = \log N$, $T = N(1 - p)$ and $\zeta_N = \frac{1}{2}W_0(\pi N^2/4)$. The value z_N of T at the crossing lies below $\zeta_N + 1/(8\zeta_N)$ and within $\zeta_N(\zeta_N + 1)/N + 1/(16\zeta_N^2)$ of it. The proof is analytic for $N \geq 175$ and uses exact integer certificates below that. The whole theorem is also proved in Lean, and the Lean kernel checks the certificates. As a consequence z_N satisfies the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with $c_0 = \frac{1}{2} \log(\pi/8)$ and an explicit bound on R_N . We also expand the crossing of each of the other bins uniformly in its distance from the edge, with a Laguerre law for fixed bins. Periodic boundaries move the central crossing by about $(\log 2)/N$ in u . Finally we prove a local law for the bin probabilities with a tilted Skellam kernel. Its relative error tends to 0 uniformly in $1 - p \leq \frac{1}{2}$ away from the edges, from the diffusive to the ballistic regime.

1 Introduction

Problem 131 of Kagey's Open Problem Collection [1] asks about a Galton board where the ball tends to keep going the same way. The first step is equally likely to be left or right, and after that each step repeats the one before it with probability p . Kagey asks when the middle bin and the endpoints are equally likely, and also about his numerator tables, the normal limit and a board on a cylinder. We answer the main question at every number of steps, and we treat every interior bin, 2 boundary conditions and the bin probabilities themselves.

Let $P_N(k)$ be the probability of exactly k right steps in N steps, with $P_0(0) = 1$. Write $q = 1 - p$, $L = \log N$ and $T = Nq$, and when $p > 0$ let $u = q/p$ be the odds of a switch. Only the 2 constant words reach the endpoints, so $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$ for $N \geq 1$. For $N \geq 2$ let p_N be the value of p at which the middle bin and the endpoints are equally likely, which is unique by Theorem 3.1, and let $z_N = N(1 - p_N)$ be the value of T there. Kagey's row n has $N = n - 1$, and his numerator tables are in Remark 2.3.

The steps of the walk form the zero-field Ising chain with free boundaries. They are the spins $\sigma_i = \pm 1$, a word has weight proportional to $\exp(J \sum_{i < N} \sigma_i \sigma_{i+1})$ with $e^{-2J} = u$, and the first spin is uniform. Antal, Droz and Rácz [5] give the law of the magnetization at a fixed scale T under several boundary conditions, and Dantchev and Rudnick [11] give exact free and periodic coefficients. The crossing happens when T is of order $\log N$, and there we need bounds at finite

N . The Bessel functions in our bounds come from the confluence of Jacobi, Laguerre and Bessel functions near the point 1 (Remark 4.5). The zeros of the row polynomial $\sum_k P_N(k)x^k$ are real for $p \leq \frac{1}{2}$, and by the Lee–Yang theorem they lie on the unit circle for $\frac{1}{2} \leq p < 1$ (Remark 2.6).

1.1 Results

Put $S_N(u) = P_N(\lfloor N/2 \rfloor)/P_N(0)$, $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$. Our first result is the uniform Bessel bound

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h} \quad (N \geq 2, x > 0)$$

of Theorem 3.3. Theorem 4.3 gives the same kind of bound for every bin, with relative error at most $Nu^2/2 + u$. These bounds hold at every N , not only in a limit. At the center they locate the crossing p_N for every N . Let $\zeta_N = \frac{1}{2}W_0(\pi N^2/4)$ be the positive root of $8\zeta e^{2\zeta} = \pi N^2$. Theorem 3.4 says that

$$\zeta_N + \frac{1}{8\zeta_N} - \frac{\zeta_N(\zeta_N + 1)}{N} - \frac{1}{16\zeta_N^2} < z_N < \zeta_N + \frac{1}{8\zeta_N} \quad \text{for every } N \geq 2.$$

The proof is analytic for $N \geq 175$. For $2 \leq N \leq 174$ it is computer-assisted, and exact integers decide the sign of $S_N - 1$ at 2 points, one on each side of the root. The whole theorem is also proved in Lean, and for $N \leq 174$ the Lean kernel checks these certificates in exact integer arithmetic. As consequences we get the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with $c_0 = \frac{1}{2} \log(\pi/8)$ and $|R_N| \leq 1/(4L^2) + (L^2 + L)/N$ for every $N \geq 2$ (Proposition 3.8), and the expansion

$$z_N = L - \frac{1}{2} \log L + c_0 + \frac{\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right)$$

of Corollary 3.7.

For a bin at distance a from the nearer edge, Theorem 4.7 expands the crossing uniformly as $a \rightarrow \infty$, with no condition on how a compares with N . For a fixed a the crossing comes instead from a Laguerre polynomial, and the odds $u_{a,b}$ at the crossing have second term $-1/b$ for every fixed bin, where $b = N - a$ (Theorem 4.8). Theorem 4.9 shows that the 2 regimes match. For odds between the scales $(\log N)/N$ of the central crossing and $N^{-1/2}$ of the adjacent one, Corollary 4.10 gives the width of the layer of bins below the endpoints.

Periodic boundaries move the central crossing in the odds by $(\log 2)/N$ to leading order (Theorem 5.2). Under both boundaries the ratio of a bin near the center to an endpoint tends to e^{t-2y^2} in the scaling of Theorem 5.3, and this counts the bins above the endpoints. Stepanyan et al. [26] study this periodic crossing, and Section 5 compares (37) with their approximations.

Finally, Section 6 writes $P_N(k)$ exactly through a difference of 2 independent binomial variables (Theorem 6.1) and approximates it by a tilted symmetric Skellam kernel. Theorem 6.2 bounds the relative error explicitly for all $0 < q < 1$, and by Corollary 6.3 the error tends to 0 uniformly in $q \leq \frac{1}{2}$ as the bin moves away from the edges, from the diffusive regime (q fixed) to the ballistic regime (T bounded). The constants are crude.

1.2 The 4 papers on Problem 131

This is the first of 4 papers on Problem 131, and Figure 1 shows the crossings of all 4. The present paper treats the walk with 2 states, which is the zero-field one-dimensional Ising chain with free

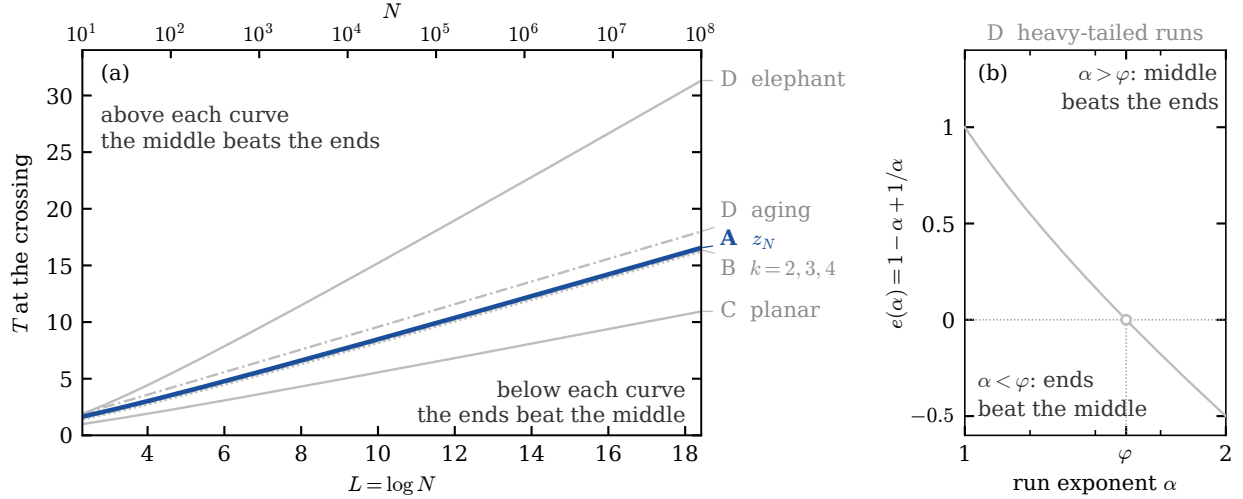


Figure 1: (a) The crossing on each model's switching scale against $L = \log N$ for Papers A to D. The scale is T for Papers A to C, $x = n\varepsilon$ for the elephant walk and the expected number of switches for the aging walk of Paper D. The bold curve is the 2-state crossing z_N of this paper, and the curves B are the crossings of Paper B for faces of $k = 2, 3, 4$ states. (b) The exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of Paper D for runs with heavy tails, which changes sign at the golden ratio φ .

boundaries. It gives explicit uniform Bessel bounds for every bin and brackets p_N for every $N \geq 2$. The bracket is analytic for $N \geq 175$ and rests on exact certificates below that, and all of it is checked in Lean. The same bounds give the crossing equation $z_N + \frac{1}{2} \log z_N = L + c_0 + 1/(8z_N) + R_N$ with an explicit bound on R_N . We also prove a uniform local law for every $q \leq \frac{1}{2}$, from the diffusive to the ballistic regime. Paper B [2] treats the complete graph, a walk on d states that moves to each other state with probability T/N . This is the one-dimensional d -state Potts chain. There every face (set of states) ties with the vertices at first order, and Paper B resolves the tie at second order. Its constants a_k are the mass of a face over a Gaussian volume, square roots of weighted spanning-tree sums, and the mode changes exactly once, from the vertices to the full centers. A start law or a weak field shifts the constants. Paper C [3] gives the general theory for a chain with transition matrix $P = I + (T/N)Q$ and $T = \tau L$. To first order the mode of the visit counts sits on a face F that minimizes $\tau \lambda_F + |F| - 1$, where λ_F is the principal Dirichlet eigenvalue of $-Q$ on F . At second order a strongly connected face has a constant K_F , and the crossings satisfy a crossing equation. The walks of this paper and of Paper B are special cases, and the application is the planar persistent walk. Paper D [4] asks what memory changes. The elephant random walk crosses at $x_n = 2z_n - \log(2\pi) + o(1)$ for even n , where z_n is the crossing of this paper at $N = n$. This is about twice the Markov value, and the profile at the crossing has 2 humps. Runs with heavy tails have a threshold at the golden ratio, and a short remark treats aging walks. The 4 papers share the notation N, L, T, z_N, λ_F and K_F .

We prove the exact law in Section 2 and the central crossing in Section 3. Section 4 treats the other bins, Section 5 periodic boundaries and the window, and Section 6 the local law. Section 7 describes the checks. The longer proofs are in Appendices A and B, and Appendix C proves the statements about Kagey's tables.

1.3 Related work

The correlated random walk goes back to Goldstein [16] and Gillis [15]. Renshaw and Henderson [25] study the same symmetric model, motivated by a Galton machine. They give the exact law from run counts, the generating function, the normal limit and the upturn at the endpoints, and in their Section 4 a Bessel limit at a fixed persistence scale, followed by a limit as that scale grows. Since the 2 limits are taken one after the other, they give no error bound at finite N when the scale grows like $\log N$, and the crossing happens at that scale. Konno [19] writes the exact law with the hypergeometric function ${}_2F_1$ and proves the telegraph limit when $p = 1 - \theta/N$ with θ fixed, which is again a fixed scale, and Herrmann and Vallois [17] give the telegraph limit with its Bessel density. Dekking and Kong [12] study the Markov binomial law, give its variance and find when it has 1, 2 or 3 modes. The OEIS entries [22, 23] already record both of Kagey's tables. Stepanyan et al. [26, Section III.B] study the periodic endpoint-adjacent and endpoint-center crossings directly, so the crossing question itself is not new. Our bounds hold at finite N when T is of order $\log N$, also near the edges. The Markov binomial estimates of Čekanavičius and Mikalauskas [7, 8] assume ranges of the transition probabilities that exclude $p \rightarrow 1$. Section 6 discusses the work on local limits.

2 The exact law

A run of the experiment with $N \geq 1$ steps is a word $w \in \{\mathbf{L}, \mathbf{R}\}^N$. Let $\text{chg}(w)$ be the number of indices $i < N$ with $w_i \neq w_{i+1}$, and let $\text{rep}(w) = N - 1 - \text{chg}(w)$. By the definition of the model,

$$\mathbb{P}(w) = \frac{1}{2} p^{\text{rep}(w)} q^{\text{chg}(w)}. \quad (1)$$

In this section we also write $\rho = 2p - 1$.

Theorem 2.1 (bin probabilities). *Let $N \geq 1$. Then $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, and for $1 \leq k \leq N - 1$*

$$P_N(k) = \frac{1}{2} \sum_{j=1}^{N-1} W(N, k, j) p^{N-1-j} q^j,$$

where, writing $a = k$ and $b = N - k$,

$$W(N, k, 2i - 1) = 2 \binom{a-1}{i-1} \binom{b-1}{i-1}, \quad W(N, k, 2i) = \binom{a-1}{i} \binom{b-1}{i-1} + \binom{a-1}{i-1} \binom{b-1}{i}$$

is the number of words with a letters $\mathbf{R}$, b letters $\mathbf{L}$ and exactly j direction changes.

Proof. By (1) it suffices to count words. A word with j changes consists of $j + 1$ maximal runs, alternately of $\mathbf{R}$'s and of $\mathbf{L}$'s. If $k \in \{0, N\}$ there is one word and it has no change. Let $a, b \geq 1$. If $j + 1 = 2i$ there are i runs of each letter and the first letter is free. The run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1} \binom{b-1}{i-1}$ such pairs. If $j + 1 = 2i + 1$ the word starts and ends with the same letter. Starting with $\mathbf{R}$ gives $i + 1$ runs of $\mathbf{R}$ and i runs of $\mathbf{L}$, so $\binom{a-1}{i} \binom{b-1}{i-1}$ words, and starting with $\mathbf{L}$ gives the other summand. $\square$

For instance $P_2 = (\frac{p}{2}, q, \frac{p}{2})$, $P_3(1) = \frac{1}{2}(1 - p^2)$ and $P_4(2) = q(1 - p + p^2)$. Note that the only word that ends in bin 0 is $\mathbf{L} \cdots \mathbf{L}$, which never switches, while each interior bin needs at least 1 switch. So the endpoints win when p is close to 1, and Section 3 asks how close.

Proposition 2.2 (recurrence and generating function). *Let $P_N^R(k)$ and $P_N^L(k)$ be the probabilities of being in bin k after N steps with last step R, respectively L, and set them to 0 outside $0 \leq k \leq N$. Then $P_1^R(1) = P_1^L(0) = \frac{1}{2}$, $P_1^R(0) = P_1^L(1) = 0$, and for $N \geq 2$*

$$P_N^R(k) = p P_{N-1}^R(k-1) + q P_{N-1}^L(k-1), \quad P_N^L(k) = p P_{N-1}^L(k) + q P_{N-1}^R(k). \quad (2)$$

Consequently

$$\sum_{N \geq 0} \sum_{k=0}^N P_N(k) x^k t^N = \frac{1 - (p - \frac{1}{2})(1+x)t}{1 - p(1+x)t + (2p-1)xt^2}, \quad (3)$$

and for all $N \geq 2$ and all k , with $P_M(k) = 0$ outside $0 \leq k \leq M$,

$$P_N(k) = p(P_{N-1}(k) + P_{N-1}(k-1)) - (2p-1)P_{N-2}(k-1). \quad (4)$$

Proof. The recursion (2) follows by conditioning on the last 2 steps. Let $H^R = \sum_{N,k} P_N^R(k) x^k t^N$ and define H^L alike. By (2) and the initial values, $H^R = xt(\frac{1}{2} + pH^R + qH^L)$ and $H^L = t(\frac{1}{2} + pH^L + qH^R)$. The determinant of this linear system is $\Delta = 1 - p(1+x)t + \rho x t^2$, since $p^2 - q^2 = \rho$. Cramer's rule gives $H^R = \frac{1}{2}xt(1 - \rho t)/\Delta$ and $H^L = \frac{1}{2}t(1 - \rho x t)/\Delta$, and $1 + H^R + H^L$ simplifies to (3). Multiplying (3) by Δ and comparing the coefficients of $x^k t^N$ with $N \geq 2$ gives (4). $\square$

At $p = \frac{1}{2}$ the last term of (4) vanishes, and (4) is Pascal's rule with weights $\frac{1}{2}$.

Remark 2.3 (numerator tables). For $p = \alpha/(\alpha + \beta)$ with integers $\alpha, \beta \geq 0$, not both 0, Kagey's numerator tables are

$$n_N(k) = 2(\alpha + \beta)^{N-1} P_N(k) \quad (N \geq 1), \quad n_0(0) = 1,$$

not reduced. By (1), $n_N(k)$ is the sum of $\alpha^{\text{rep}(w)} \beta^{\text{chg}(w)}$ over the words with k letters R and $N - k$ letters L, so it is an integer, and by (4) $n_N(k) = \alpha(n_{N-1}(k) + n_{N-1}(k-1)) - (\alpha^2 - \beta^2)n_{N-2}(k-1)$ for $N \geq 3$. If we read a word as a monotone lattice path to $(k, N - k)$, then $n_N(k)$ counts these paths with each straight continuation colored in one of α ways and each corner in one of β ways. For $p = \frac{2}{3}$ the table counts rook paths (A035002 read by antidiagonals), and for $p = \frac{1}{3}$ it counts Mathar's blocked walks (A348595). The OEIS entries already record the generating functions (by Kruchinin, Somos and Mathar) [22, 23], and our bijections explain them. The proofs are in Appendix C, and these identities are also proved in Lean (Section 7).

2.1 Variance and the normal limit

Let $\sigma_i = +1$ if step i is R and $\sigma_i = -1$ otherwise, as in the introduction, and put $X_N = \sigma_1 + \dots + \sigma_N$ and $K_N = (N + X_N)/2$ for the bin. Then $\sigma_{i+1} = \sigma_i s_i$, where $s_1, s_2, \dots$ are independent of each other and of σ_1 , with $\mathbb{P}(s_i = 1) = p$ and $\mathbb{P}(s_i = -1) = q$. So $\mathbb{E}s_i = \rho$.

Proposition 2.4 (mean and variance). $\mathbb{E}K_N = N/2$, and for $p < 1$

$$\text{Var } K_N = \frac{1}{4} \left(N \frac{1 + \rho}{1 - \rho} - \frac{2\rho(1 - \rho^N)}{(1 - \rho)^2} \right) = \frac{Np}{4(1 - p)} - \frac{(2p - 1)(1 - (2p - 1)^N)}{8(1 - p)^2}.$$

For $p = 1$, $\text{Var } K_N = N^2/4$.

Proof. Note that $\mathbb{E}\sigma_1 = 0$ and $\sigma_{i+m} = \sigma_i s_i \dots s_{i+m-1}$, so $\mathbb{E}\sigma_i = 0$ and $\mathbb{E}\sigma_i \sigma_{i+m} = \rho^m$. Hence, $\text{Var } X_N = N + 2 \sum_{m=1}^{N-1} (N - m) \rho^m$. Summing the 2 geometric series gives the first form, and the second follows from $(1 + \rho)/(1 - \rho) = p/q$ and $1 - \rho = 2q$. $\square$

Theorem 2.5 (normal limit). *Let $0 < p < 1$. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution to the normal law with mean 0 and variance $p/(4(1-p))$.*

Proof. Let $\varphi_N(\theta) = \mathbb{E}e^{i\theta X_N}$, and let v_N be the vector with entries $\mathbb{E}[e^{i\theta X_N}; \sigma_N = \pm 1]$. By (2), $v_{N+1} = Mv_N$ with

$$M = \begin{pmatrix} pe^{i\theta} & qe^{i\theta} \\ qe^{-i\theta} & pe^{-i\theta} \end{pmatrix}, \quad \text{tr } M = 2p \cos \theta, \quad \det M = \rho.$$

The eigenvalues are $\lambda_{\pm}(\theta) = p \cos \theta \pm \sqrt{q^2 - p^2 \sin^2 \theta}$. For $|\sin \theta| < q/p$ they are real and distinct, so $\varphi_N = c_+ \lambda_+^{N-1} + c_- \lambda_-^{N-1}$ for $N \geq 1$. The values $\varphi_1 = \cos \theta$ and $\varphi_2 = p \cos 2\theta + q$ give $c_+ = (\varphi_2 - \lambda_- \cos \theta)/(\lambda_+ - \lambda_-)$ and $c_- = \cos \theta - c_+$. As $\theta \rightarrow 0$ we have $\lambda_- \rightarrow \rho$, $c_+ \rightarrow 1$, $c_- \rightarrow 0$ and

$$\lambda_+(\theta) = p\left(1 - \frac{\theta^2}{2}\right) + q\left(1 - \frac{p^2 \theta^2}{2q^2}\right) + O(\theta^4) = 1 - \frac{p}{2q} \theta^2 + O(\theta^4).$$

Put $\theta = s/\sqrt{N}$ with s fixed. Since $|\rho| < 1$, we get $\varphi_N(s/\sqrt{N}) \rightarrow \exp(-\frac{p}{2q}s^2)$. By Lévy's continuity theorem [6, Theorem 26.3], $X_N/\sqrt{N}$ converges to the normal law with variance p/q . Finally $K_N - N/2 = X_N/2$. $\square$

For $p = 1$ the walk ends in bin 0 or bin N , and for $p = 0$ in a middle bin, so these 2 cases have no normal limit.

Remark 2.6 (zeros of the generating polynomial). Let $N \geq 2$ and $g_N(x) = \sum_k P_N(k)x^k$. First let $0 < p \leq \frac{1}{2}$. By (4), $g_N = p(1+x)g_{N-1} + (1-2p)xg_{N-2}$ with a nonnegative last coefficient. Then every g_N has only real zeros, as in the proof of [12, Proposition 4.2], and they are negative since g_N has positive coefficients. By a classical result recalled by Pitman [24, Proposition 1], K_N then has the law of a sum of N independent Bernoulli variables. So each row is log-concave and no bin is less likely than the endpoints, which fits with Section 4, where every endpoint crossing is at least $\frac{2}{3}$.

Now let $\frac{1}{2} \leq p < 1$, so that $u \leq 1$. Put $x_{i,i+1} = x_{i+1,i} = u$ for $1 \leq i < N$ and $x_{ij} = 1$ for the other pairs $i \neq j$. For $S \subseteq \{1, \dots, N\}$ let w be the word with R exactly at the positions in S . Each switch of w is a pair of neighbors split by S , so $\prod_{i \in S, j \notin S} x_{ij} = u^{\text{chg}(w)}$, and (1) gives the lattice-gas form

$$g_N(x) = \frac{p^{N-1}}{2} \sum_{S \subseteq \{1, \dots, N\}} x^{|S|} \prod_{i \in S, j \notin S} x_{ij}.$$

The x_{ij} are real and symmetric with $|x_{ij}| \leq 1$, so by the Lee–Yang circle theorem [20], in the form quoted by Ghosh [14], every zero of g_N lies on the circle $|x| = 1$. It applies because the boundaries are free, so only the $N-1$ bonds of the path carry a factor u , and the first step is uniform, so there is no boundary field. Also $u = e^{-2J} \leq 1$ means $J \geq 0$, which holds exactly when $p \geq \frac{1}{2}$.

At $p = \frac{1}{2}$ we have $g_N = ((1+x)/2)^N$, and both statements hold. For $\frac{1}{2} < p < 1$ the zeros are never all real. Otherwise K_N would again have such a law, with variance at most $N/4$. But the proof of Proposition 2.4 gives $\text{Var } X_N = N + 2 \sum_{m=1}^{N-1} (N-m)\rho^m > N$, so $\text{Var } K_N > N/4$.

Remark 2.7 (cylinder). On a cylinder whose rows have c pegs the local rule is unchanged, so the position is X_N modulo $2c$ in half-peg units. The Fourier coefficients of its law are $\varphi_N(\pi l/c)$, $0 \leq l < 2c$, equal to 1 and $(-1)^N$ at $l = 0$ and $l = c$. For $0 < p < 1$ and $\theta \notin \pi\mathbb{Z}$ both eigenvalues of M have modulus less than 1, since real ones satisfy $|\lambda_{\pm}| \leq p|\cos \theta| + q$ and complex ones have modulus $\sqrt{\rho}$. So $\varphi_N(\theta) \rightarrow 0$ geometrically, also for a double eigenvalue $\lambda = p \cos \theta$, since then

$(M - \lambda I)^2 = 0$ by the Cayley–Hamilton theorem and $M^n = \lambda^n I + n\lambda^{n-1}(M - \lambda I)$. Hence the law tends geometrically fast to the uniform law on the c positions of the correct parity, with no normal shape in the limit.

3 The central crossing at every N

Let $N \geq 2$ and let $m_N = \lfloor N/2 \rfloor$ be a middle bin (for odd N the 2 middle bins have equal probability by symmetry). Put $u = q/p$. By Theorem 2.1,

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j) u^j, \quad (5)$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$.

Each switch in a word gives a factor of u , and a word that ends in the middle can switch almost anywhere, so $S_N(u)$ is roughly e^{Nu}/N . So we expect the crossing where $Nu \approx L$.

Theorem 3.1. (a) For every $N \geq 2$ there is exactly one $p = p_N \in [0, 1]$ with $P_N(m_N) = P_N(0)$, and $p_N \geq \frac{2}{3}$. The middle is more likely than the endpoints for $p < p_N$ and less likely for $p > p_N$.

(b) $p_2 = \frac{2}{3}$, $p_3 = 1/\sqrt{2}$, p_4 is the real root of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.

(c) $p_N \rightarrow 1$, and more precisely $1 - p_N \sim (\log N)/N$.

Proof. (a) At $p = 0$ the endpoints have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (5). S_N is continuous and strictly increasing on $[0, \infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$, so there is exactly one root $u_N \in (0, \frac{1}{2}]$, and $p = 1/(1 + u)$ is a decreasing bijection from $[0, \infty)$ to $(0, 1]$. Hence $p_N = 1/(1 + u_N) \geq \frac{2}{3}$.

(b) $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$ give the 3 values. For p_4 , substitute $u = (1 - p)/p$ in $S_4(u) = 1$. For monotonicity we compare coefficients. Going from N to $N + 1$, one entry of the middle pair $(a, b) = (\lfloor N/2 \rfloor, \lceil N/2 \rceil)$ grows by 1 and the other is unchanged. The formulas of Theorem 2.1 are symmetric in (a, b) , and every $\binom{a-1}{\cdot}$ is nondecreasing in a . So $W(N + 1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j . The coefficient of u^2 is $a + b - 2 = N - 2$, which increases strictly. Hence, $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) By Corollary 3.7(b) below, used only for $N \geq 175$ where its proof is analytic, $z_N = L + O(\log L)$, which gives (c). $\square$

In Kagey's terms, the middle and the endpoints of row $N + 1$ are level at $p = p_N$. The least value is $p_2 = \frac{2}{3}$ for the odd rows, which is why the third row of his $p = \frac{2}{3}$ table is 2, 2, 2, and $p_3 = 1/\sqrt{2}$ for the even rows after the second (which is 1, 1 for every p). There is no greatest value, since $p_N \uparrow 1$. The tenth row has $p_9 = 0.82829\dots$, just below $\frac{5}{6}$, so its middle and endpoints are nearly level in his third histogram (Figure 2 and Table 1).

The Bessel functions come in because the runs are long. When Nu is about L , a typical word only switches about L times. Then the number of ways to split a letters into $r + 1$ runs, $\binom{a-1}{r}$, is close to $a^r/r!$, and after this change the run count is exactly the power series of I_0 and I_1 . The next lemma controls the error of this change.

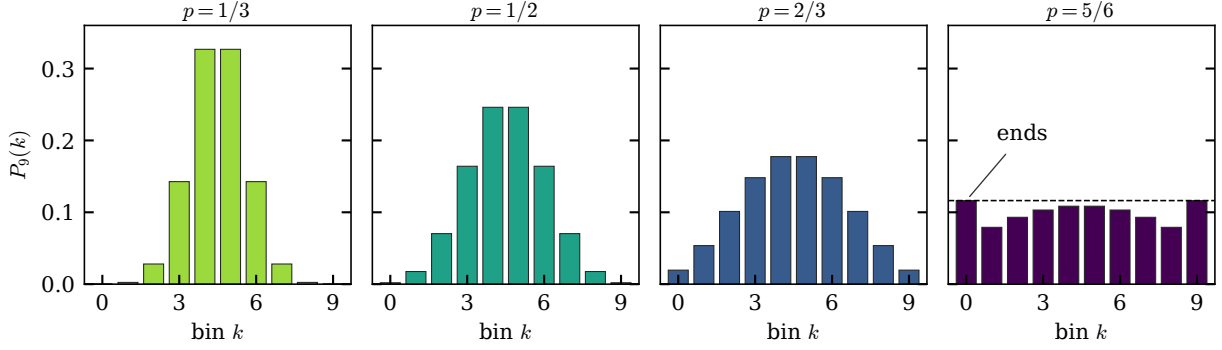


Figure 2: The bin probabilities $P_9(k)$ of Kagey's tenth row for $p = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$. At $p = \frac{5}{6}$, just above p_9 , the 2 middle bins are slightly below the endpoints (dashed line).

N	p_N	N	p_N	N	p_N	N	p_N
2	0.66666667	6	0.78871211	10	0.83840411	50	0.94200405
3	0.70710678	7	0.80361731	11	0.84652725	100	0.96496633
4	0.74487450	8	0.81759938	12	0.85426060	200	0.97939018
5	0.76759188	9	0.82829163	20	0.89293837	400	0.98812464

Table 1: The value p_N at which the middle and endpoint bins of row $N + 1$ are equally likely.

Lemma 3.2 (clipped products). (a) If $t_1, \dots, t_r \geq 0$ and $(v)_+ = \max(v, 0)$, then $0 \leq 1 - \prod_{k=1}^r (1 - t_k)_+ \leq \sum_{k=1}^r t_k$. (b) For integers $m \geq 1$ and $r \geq 0$, $U_r(m) := \frac{r!}{m^r} \binom{m-1}{r} = \prod_{k=1}^r (1 - \frac{k}{m})_+$, so $0 \leq 1 - U_r(m) \leq \frac{r(r+1)}{2m}$. (c) For $x \geq 0$,

$$\sum_{r \geq 0} \frac{r(r+1)x^{2r}}{(r!)^2} = x^2 I_0(2x) + x I_1(2x),$$

$$\sum_{r \geq 0} \frac{r(r+1)x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x), \quad \sum_{r \geq 0} \frac{(r+1)x^{2r+1}}{r!(r+1)!} = x I_0(2x).$$

Proof. (a) Every factor lies in $[0, 1]$, so the product is at most 1. If some $t_k \geq 1$ the product is 0 and the sum is at least 1. Otherwise $\prod_k (1 - t_k) \geq 1 - \sum_k t_k$ by induction on r . (b) Note that $\frac{r!}{m^r} \binom{m-1}{r} = \prod_{k=1}^r \frac{m-k}{m}$. For $r < m$ every factor is positive, and for $r \geq m$ both sides vanish, the right side through the factor $k = m$. The bound is (a) with $t_k = k/m$. (c) We use $I_0(2x) = \sum_r x^{2r}/(r!)^2$ and $I_1(2x) = \sum_r x^{2r+1}/(r!(r+1)!)$ [28, Eq. 10.25.2]. Write $r(r+1) = r^2 + r$ in the first sum, cancel the factors r and $r+1$ against the factorials, and shift the index. $\square$

Theorem 3.3 (uniform Bessel approximation). Put $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$, and let $u_N = (1 - p_N)/p_N$.

(i) For every $N \geq 2$ and $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}. \quad (6)$$

(ii) Let $x_N = hu_N$, let x_B be the unique positive root of $F(x_B) = h$, and put $\delta_N = x_N(x_N + 1)/h$. Then $x_B \leq x_N$, and $x_N - x_B \leq -\frac{1}{2} \log(1 - \delta_N)$ whenever $\delta_N < 1$.

(iii) With $u_B = x_B/h$ we have $0 \leq u_N - u_B = O(L^2/N^2)$.

Proof. (i) Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$. As in Lemma 3.2(b), $A_r = \frac{r!}{h^r} \binom{a-1}{r} = \prod_{k=1}^r (1 - \frac{h-a+k}{h})_+$, and B_r is the same with b in place of a . Since $a+b = 2h$, the $2r$ numbers $(h-a+k)/h$ and $(h-b+k)/h$ are nonnegative with sum $r(r+1)/h$, so Lemma 3.2(a) gives $0 \leq 1 - A_r B_r \leq r(r+1)/h$. In the same way the deficits of $A_{r+1} B_r$ and $A_r B_{r+1}$ are at most $((r+1)^2 + h - a)/h$ and $((r+1)^2 + h - b)/h$. Averaging cancels the parity offsets, so $0 \leq 1 - \frac{1}{2}(A_{r+1} B_r + A_r B_{r+1}) \leq (r+1)^2/h$. With $u = x/h$ the run-count formula of Theorem 2.1 becomes

$$S_N(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} A_r B_r + \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} \frac{A_{r+1} B_r + A_r B_{r+1}}{2} \right].$$

Without the factors A_r, B_r the bracket is $I_0(2x) + I_1(2x)$ and the right side is $F(x)/h$. By the deficit bounds and Lemma 3.2(c), $F(x)/h - S_N(u)$ lies between 0 and $2u/h$ times $\sum_r r(r+1) \frac{x^{2r}}{(r!)^2} + \sum_r (r+1)^2 \frac{x^{2r+1}}{r!(r+1)!} = x(x+1)(I_0(2x) + I_1(2x))$. This proves (i), including the tail beyond the finite support of the polynomial.

(ii) Differentiating the series term by term gives $F' = 2F + 2I_0(2x)$. So F increases from 0 to ∞ , x_B is unique, and $(\log F)' \geq 2$. At $x = x_N$ we have $hS_N(u_N) = h$, so (i) gives $h \leq F(x_N)$, hence $x_B \leq x_N$, and $F(x_N) \leq h/(1 - \delta_N)$ when $\delta_N < 1$. Integrating $(\log F)' \geq 2$ from x_B to x_N gives $2(x_N - x_B) \leq \log(F(x_N)/h) \leq -\log(1 - \delta_N)$.

(iii) By Theorem 3.1(c), whose proof does not use (iii), $x_N \sim L/2$. So $\delta_N = O(L^2/N)$, and (ii) gives $x_N - x_B = O(L^2/N)$. Divide by h . $\square$

The bound (6) and parts (ii) and (iii) are proved in Lean, in the files `UniformBessel.lean`, `RootBounds.lean` and `Asymptotics.lean` of `lean/PaperA`.

3.1 The crossing at every N

We measure the crossing by the value z_N of $T = Nq$ at $p = p_N$, as in the introduction. In terms of the odds,

$$z_N = \frac{Nu_N}{1 + u_N}. \quad (7)$$

Let $\psi(z) = z + \frac{1}{2} \log z$ and $c_0 = \frac{1}{2} \log(\pi/8) = -0.46735\dots$, and let ζ_N be the positive root of

$$\psi(\zeta_N) = L + c_0, \quad \text{that is,} \quad 8\zeta_N e^{2\zeta_N} = \pi N^2 \quad \text{or} \quad \zeta_N = \frac{1}{2} W_0(\pi N^2/4), \quad (8)$$

with W_0 the principal branch of the Lambert function. Since $\zeta e^{2\zeta}$ increases, ζ_N increases with N . Also $\zeta_N \leq L$, since $\psi(L) \geq L + c_0$ when $L \geq \pi/8$. Finally let $\varepsilon_N = \zeta_N(\zeta_N + 1)/N + 1/(16\zeta_N^2)$.

Theorem 3.4 (the crossing at every N). *For every $N \geq 2$,*

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N}.$$

The proof is analytic for $N \geq 175$ and computer-assisted for $2 \leq N \leq 174$. Since $\zeta_N \leq L$ and $\zeta_N \rightarrow \infty$, the width ε_N tends to 0. It is 0.440, 0.163, 0.0387 and 0.00778 at $N = 10, 10^2, 10^3, 10^4$ (Table 2).

Let $\Phi(w) = F(w/2) = w(I_0(w) + I_1(w))$ and $E(z) = \sqrt{\pi z/2} e^{-z}(I_0(z) + I_1(z))$, and put $w_N = Nu_N$ and $\Lambda_N = 2\Phi(w_N)/N$. At the crossing, (6) and (9) below give $\psi(w_N) + \log E(w_N) = L + c_0 + \log \Lambda_N$ with $0 \leq \log \Lambda_N \leq -\log(1 - w_N(w_N + 2)/(2N))$. Since $\log E(w) = -1/(8w) + O(w^{-2})$ and $z_N = w_N + O(w_N^2/N)$, this suggests $z_N = \zeta_N + 1/(8\zeta_N) + O(\zeta_N^{-2} + \zeta_N^2/N)$. The theorem makes this error explicit and one-sided.

Lemma 3.5 (root bracket). *Let $N \geq 2$ and $w > 0$. Then*

$$\log \Phi(w) - \log \frac{N}{2} = \psi(w) - \psi(\zeta_N) + \log E(w). \quad (9)$$

(a) *If $\Phi(w) \leq N/2$, then $Nw/(N+w) \leq z_N$, strictly if $\Phi(w) < N/2$. (b) *If $\Phi(w)(1 - \frac{w(w+2)}{2N}) > N/2$, then $z_N < Nw/(N+w)$.**

Proof. Note that $\log \Phi(w) = \psi(w) + \frac{1}{2} \log \frac{2}{\pi} + \log E(w)$ and $\log \frac{N}{2} - \psi(\zeta_N) = -\log 2 - c_0 = \frac{1}{2} \log \frac{2}{\pi}$, which gives (9). Now apply (6) with $x = w/2$. Then $F(x) = \Phi(w)$ and $x(x+1)/h = w(w+2)/(2N)$, so

$$\Phi(w) \left(1 - \frac{w(w+2)}{2N}\right) \leq \frac{N}{2} S_N\left(\frac{w}{N}\right) \leq \Phi(w).$$

(a) The hypothesis gives $S_N(w/N) \leq 1 = S_N(u_N)$, strictly if $\Phi(w) < N/2$. Since S_N is strictly increasing, $w/N \leq u_N$, strictly in that case. The map $v \mapsto Nv/(1+v)$ increases and sends w/N to $Nw/(N+w)$ and u_N to z_N . (b) Now $S_N(w/N) > 1$, so $w/N > u_N$. $\square$

Lemma 3.6 (Bessel bounds). *For every $z > 0$,*

$$1 - \frac{1}{8z} - \frac{3}{128z^2} - \frac{45}{512z^3} \leq E(z) \leq 1 - \frac{1}{8z} - \frac{3}{128z^2} + \frac{15}{512z^3},$$

and $E(z) \leq 1$. In particular $E(z) \leq 1 - \frac{1}{8z}$ for $z \geq \frac{5}{4}$.

At the center this lemma replaces the large-argument expansion of $I_0 + I_1$, with explicit errors. By [28, Eq. 10.32.3] and a substitution, $E(z) = \pi^{-1/2} \int_0^{2z} e^{-v} v^{-1/2} \sqrt{1-v/(2z)} dv$. We bound $\sqrt{1-y}$ above and below by cubics that share its Taylor terms up to y^2 , extend the integral to ∞ , and use $\Gamma(k + \frac{1}{2})/\Gamma(\frac{1}{2}) = 1, \frac{1}{2}, \frac{3}{4}, \frac{15}{8}$ for $k = 0, 1, 2, 3$. The proof is in Appendix A.

Proof of Theorem 3.4. Since $N = \sqrt{8\zeta_N/\pi} e^{\zeta_N}$ and $\sqrt{32/\pi} e^4 = 174.25\dots$, we have $\zeta_N \geq 4$ exactly when $N \geq 175$. Write $\zeta = \zeta_N$ and $U = \zeta + \frac{1}{8\zeta}$.

Let $N \geq 175$. For the upper bound we apply Lemma 3.5(b) at $W = NU/(N-U)$, for which $NW/(N+W) = U$. By (9) it suffices that $G_U = \psi(W) - \psi(\zeta) + \log E(W) + \log(1 - W(W+2)/(2N)) > 0$, and Lemma 3.6 gives $G_U \geq \frac{1}{32\zeta^2} + \frac{\zeta(\zeta-1)}{2N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$. For the lower bound we apply Lemma 3.5(a) at $W' = N\ell/(N-\ell)$ with $\ell = U - \varepsilon_N$. It suffices that $G_\ell = \psi(W') - \psi(\zeta) + \log E(W') < 0$, and we get $G_\ell \leq -\frac{3}{128\zeta^2} - \frac{\zeta}{N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$. The explicit constants for every $\zeta \geq 4$ are in Appendix A.

For $2 \leq N \leq 174$ the proof is computer-assisted. Let $d = 10^9$ and let $c < c'$ be integers with $c' = c + 1$ ($c' = c + 2$ at $N = 2$, where $u_2 = \frac{1}{2}$ is a grid point). The sign of $S_N(c/d) - 1$ is the sign of $n_N(m_N) - d^{N-1}$, where $n_N(m_N)$ is the numerator of Proposition C.1 (Appendix C) at $(\alpha, \beta) = (d, c)$, by the identity in Appendix A. We computed these integers at c and c' by 2 exact methods, the run-count sum of Theorem 2.1 and the recursion (2) with integer weights d, c in place of p, q . They agreed for every $N \leq 174$ and gave $c/d < u_N < c'/d$, so $z(c/d) < z_N < z(c'/d)$ with $z(u) = Nu/(1+u)$. We then checked $\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z(c/d)$ and $z(c'/d) < \zeta_N + \frac{1}{8\zeta_N}$ in 60-digit decimal arithmetic, with ζ_N known to within 10^{-45} . We certified this bracket by the sign of $\psi(\zeta) - L - c_0$, which is the sign of $8\zeta e^{2\zeta} - \pi N^2$, at both of its ends, using a correctly rounded logarithm. The smallest margins are 0.033193 (upper) and 0.085591 (lower), both at $N = 174$, so every margin exceeds 0.03. The Lean proof checks the same comparisons with rigorous rational bounds on ζ_N and π . The certificates are in `data/every_n_certificates.csv`, made by `verify_every_n.py`. As a cross-check they pass for every $N \leq 1000$ (with both methods for $N \leq 400$ and the run-count sum alone above). $\square$

The theorem is also proved in Lean for every $N \geq 2$, together with Lemmas 3.5 and 3.6, in the files `BesselIntegral.lean`, `EveryN.lean` and `EveryNLarge.lean` of `lean/PaperA`. For $N \leq 174$ the Lean kernel checks the same certificates in exact integer arithmetic (Section 7).

N	z_N	ζ_N	$z_N - \zeta_N - \frac{1}{8\zeta_N}$	$-\varepsilon_N$	R_N
2	0.6666666666667	0.5368290974	-0.1030112	-0.62938	+0.050643
10	1.615958871376	1.6001732978	-0.0623310	-0.44048	-0.056660
100	3.503367368092	3.5100055489	-0.0422507	-0.16337	-0.043265
174	3.996779704730	3.9987131703	-0.0331935	-0.11878	-0.033450
175	4.001967989170	4.0038072816	-0.0330596	-0.11838	-0.033304
1000	5.590351166975	5.5807388611	-0.0127862	-0.03873	-0.011887
10^4	7.734265514999	7.7210118331	-0.0029359	-0.00778	-0.002051

Table 2: The crossing against Theorem 3.4, and the remainder R_N of (12). We computed z_N by run-count bisection in 60-digit decimal arithmetic and by a floating-point dynamic program, which agree to the digits shown.

Corollary 3.7. Put $\nu_N = L - \frac{1}{2} \log L + c_0$.

(a) For every $N \geq 2$, $1 - (\zeta_N + \frac{1}{8\zeta_N})/N < p_N < 1 - (\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N)/N$.

(b) For every $N \geq 2$, $\nu_N \leq \zeta_N \leq \nu_N + \frac{\log L - 2c_0}{4\nu_N}$ and

$$-\varepsilon_N < z_N - \nu_N < \frac{\log L - 2c_0 + \frac{1}{2}}{4\nu_N}. \quad (10)$$

In particular $z_N = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$.

(c) As $N \rightarrow \infty$,

$$z_N = \nu_N + \frac{\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (11)$$

(d) $z_N > \zeta_N$ for every $N \geq 2774$.

Proof. (a) is Theorem 3.4 with $p_N = 1 - z_N/N$. (b) Note that $\zeta_N = L + c_0 - \frac{1}{2} \log \zeta_N$. Since $\zeta_N \leq L$, this gives $\zeta_N \geq \nu_N$, and $\nu_N > 0$ since it increases in N and $\nu_2 = 0.41$. Also $\zeta_N - \nu_N = \frac{1}{2} \log(L/\zeta_N) \leq (L - \zeta_N)/(2\zeta_N)$ and $L - \zeta_N = \frac{1}{2} \log \zeta_N - c_0 \leq \frac{1}{2} \log L - c_0$. Hence $\zeta_N - \nu_N \leq (\log L - 2c_0)/(4\nu_N)$. For z_N , add Theorem 3.4 and $0 < \frac{1}{8\zeta_N} \leq \frac{1}{8\nu_N}$. (c) Put $\eta = \zeta_N - \nu_N = \frac{1}{2} \log(L/\zeta_N)$. By (b), $L - \zeta_N = \frac{1}{2} \log L - c_0 + O(\log L/L)$, so $\eta = \frac{L - \zeta_N}{2L} + O\left(\frac{(\log L)^2}{L^2}\right) = \frac{\frac{1}{4} \log L - \frac{1}{2} c_0}{L} + O\left(\frac{(\log L)^2}{L^2}\right)$. Also $\frac{1}{8\zeta_N} = \frac{1}{8L} + O\left(\frac{\log L}{L^2}\right)$ and $\varepsilon_N = O(L^2/N + L^{-2})$, and Theorem 3.4 gives (c). (d) By Theorem 3.4, $z_N - \zeta_N > \frac{1}{8\zeta_N} - \varepsilon_N$, which is positive when $f(\zeta_N) > 1$, where $f(\zeta) = n(\zeta)(\frac{1}{8\zeta} - \frac{1}{16\zeta^2})/(\zeta(\zeta+1))$ and $n(\zeta) = \sqrt{8\zeta/\pi} e^\zeta$, so that $n(\zeta_N) = N$. The logarithmic derivative of f is $1 + \frac{1}{2\zeta} - \frac{2}{\zeta} + \frac{1}{2\zeta^2 - \zeta} - \frac{1}{\zeta+1} > 1 - \frac{3}{2\zeta} - \frac{1}{\zeta+1} > 0$ for $\zeta \geq 3$, so once $f(\zeta_N) > 1$ it stays above 1. This step is computer-assisted. In 60-digit decimal arithmetic, with ζ_N bracketed within 10^{-45} as in the proof of Theorem 3.4, we found $f(\zeta_{2774}) - 1 = 2.1 \times 10^{-4}$, and the sign holds on the whole bracket (`verify_every_n.py` and `verify_every_n_guards.py`). The condition fails at $N = 2773$. $\square$

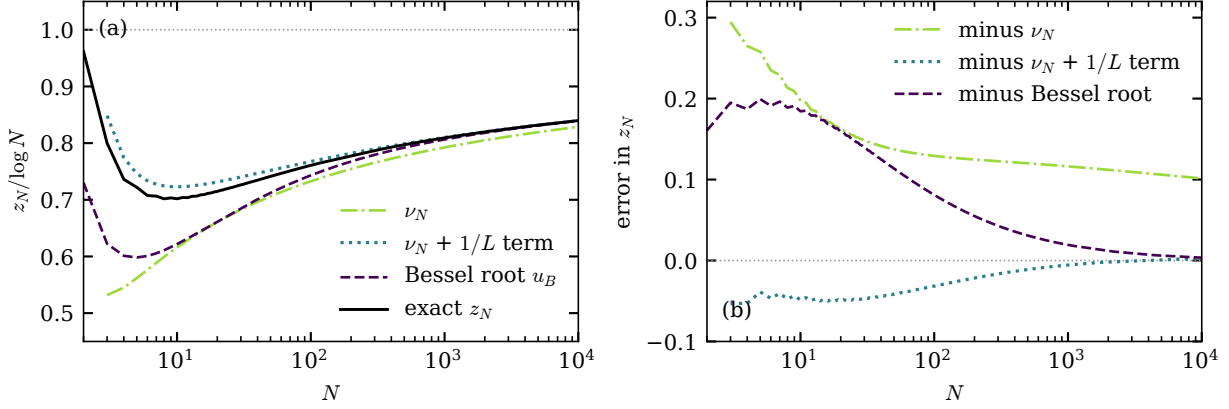


Figure 3: (a) The scaled crossing $z_N / \log N$ (solid) and the same quantity from the Bessel root u_B of Theorem 3.3 (dashed), from ν_N of (10) (through the constant) and from $\nu_N + (\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}) / L$ of (11) (with the $1/L$ term). The ratio tends to 1 very slowly. (b) The exact z_N minus each approximation.

Proposition 3.8 (the crossing equation). *For every $N \geq 2$,*

$$z_N + \frac{1}{2} \log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \quad -\frac{z_N^2 + z_N}{N} \leq R_N \leq \frac{1}{16z_N^2}. \quad (12)$$

Moreover $L/2 \leq z_N \leq L$, so $|R_N| \leq \frac{1}{4L^2} + \frac{L^2 + L}{N}$ for every $N \geq 2$.

Paper C [3] gets this equation with an error $O(1/L)$ in place of $1/(8z_N) + R_N$, as a special case of its crossing equation, and Paper D [4] compares the elephant walk with it. For the proof, recall $\Lambda_N = 2\Phi(w_N)/N$ and let $\delta = w_N(w_N + 2)/(2N)$. By (6), $1 \leq \Lambda_N \leq 1/(1 - \delta)$ when $\delta < 1$, and (9) turns the definition of R_N into the exact formula (45). For $N \geq 175$ the proof is analytic. Lemma 3.6 puts the Bessel part of (45) between 0 and $0.0559/z_N^2$, and the other terms lie between $-(z_N^2 + z_N)/N$ and 0. For $N \leq 174$ the proof is computer-assisted and uses the same certificates, and $z_2 = \frac{2}{3} < \log 2$ holds by hand since it means $e^2 < 8$. The details are in Appendix A.

Remark 3.9 (the sign of $z_N - \zeta_N$, computer-assisted). We have $z_N > \zeta_N$ for $N \leq 15$, $z_N < \zeta_N$ for $16 \leq N \leq 218$, and $z_N > \zeta_N$ for every $N \geq 219$. For $N \leq 1000$ the certificates decide the sign, since ζ_N never lies in $[z(c/d), z(c'/d)]$. The smallest distance is 1.66×10^{-5} , at $N = 218$. Both exact methods cover $N \leq 400$, and the run-count sum alone covers the rest. For $1001 \leq N \leq 2773$ we checked $S_N(v) < 1$ in exact integers from the run-count sum (`verify_every_n_sign.py`), at a rational v above $\zeta_N/(N - \zeta_N)$ by at least 1.2×10^{-11} , where $z(\zeta_N/(N - \zeta_N)) = \zeta_N$. These integers were cross-checked by floating-point run-count and dynamic-program sign tests for every $N \leq 3000$, and by the integer recursion at $N = 401, 1000, 1008, 2000, 2773$. For $N \geq 2774$ the sign is Corollary 3.7(d).

The ratios u_B/u_N are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$. We predicted intervals for the last 3 before computing them (`data/ledger.md`), and Theorem 3.3 proves the approximation behind those predictions. Figure 3 compares the exact crossing with these approximations up to $N = 10^4$.

4 The full hierarchy of endpoint crossings

Let $a, b \geq 1$ be integers, $N = a + b$, $u = (1 - p)/p$ with $p > 0$, and $S_{a,b}(u) = P_N(a)/P_N(0)$. Theorem 2.1 and $\binom{m-1}{r+1} = \frac{m-1-r}{r+1} \binom{m-1}{r}$ for $m = a, b$ give

$$S_{a,b}(u) = \sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r} \left(2u^{2r+1} + \frac{N-2-2r}{r+1} u^{2r+2} \right). \quad (13)$$

Its coefficients are nonnegative, as $N-2-2r \geq |b-a|$ for $r < \min(a, b)$, and the linear one is 2. So there is a unique $u_{a,b} > 0$ with $S_{a,b}(u_{a,b}) = 1$, and we put $p_{a,b} = 1/(1 + u_{a,b})$. As $S_{a,b}(u) \geq 2u$, $u_{a,b} \leq \frac{1}{2}$ and $p_{a,b} \in [\frac{2}{3}, 1)$, with $p_{a,b} = \frac{2}{3}$ only for $a = b = 1$.

Proposition 4.1 (ordering the crossings). *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \quad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b . In fact, the corresponding inequalities between the polynomials $S_{a+1,b-1}$ and $S_{a,b}$, or $S_{a,b+1}$ and $S_{a,b}$, hold coefficientwise, with at least one strict coefficient.

Proof. For fixed N and $r \leq a - 1$, $\binom{a-1}{r} \binom{b-1}{r} = (r!)^{-2} \prod_{j=0}^{r-1} (a-1-j)(b-1-j)$, and each factor grows when (a, b) becomes $(a+1, b-1)$, by $(a-j)(b-2-j) - (a-1-j)(b-1-j) = b-a-1 > 0$. The even multiplier $(N-2-2r)/(r+1)$ is unchanged, and nonnegative where either product is nonzero, since then $r \leq a$ and $N-2-2r \geq b-a-2 \geq 0$. New summands are nonnegative, and the cubic coefficient grows strictly. Increasing b with a fixed increases the binomial coefficients of Theorem 2.1, and the quadratic coefficient $N-2$ strictly. Comparing the unique roots proves the rest. $\square$

The ordering of the bin heights below also follows from the interior log-concavity of Dekking and Kong [12, Proposition 4.3]. We use the coefficients.

Proposition 4.2 (adjacent bins and global extrema). *For $N \geq 2$ and $0 < p < 1$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\text{adj}} = \frac{1 + \sqrt{N-1}}{2 + \sqrt{N-1}}.$$

For $0 < p < 1$, the endpoints are the least likely bins if $p < p_N^{\text{adj}}$, bins 1 and $N-1$ are if $p > p_N^{\text{adj}}$, and they tie at equality. For $N \geq 4$, the central crossing satisfies $p_N > p_N^{\text{adj}}$. At $p = p_N$, the middle and endpoints attain the global maximum and bins 1, $N-1$ attain the global minimum. For $N = 2, 3$, $p_N = p_N^{\text{adj}}$ and every bin ties at the crossing.

Proof. By (13), $S_{1,N-1}(u) = 2u + (N-2)u^2$, with positive root $u^{\text{adj}} = 1/(1 + \sqrt{N-1})$. This gives p_N^{adj} and the sign change. For $0 < p < 1$ we have $P_N(k) = P_N(0)S_{k,N-k}(u)$ with $P_N(0) > 0$, so if $1 \leq k < N-k-1$ Proposition 4.1 gives $P_N(k) < P_N(k+1)$. By symmetry the middle is the largest interior bin and bins 1, $N-1$ are the smallest, which gives the claims about the global minimum. For $N \geq 4$ the cubic coefficient $2(\lfloor N/2 \rfloor - 1)(\lceil N/2 \rceil - 1)$ of S_N is positive, so $2u_N + (N-2)u_N^2 < S_N(u_N) = 1$, hence $u_N < u^{\text{adj}}$ and $p_N > p_N^{\text{adj}}$. At p_N the middle ties with the endpoints, which gives the global maximum. For $N = 2, 3$ we have $S_N = S_{1,N-1}$, and symmetry gives the ties. $\square$

So $1 - p_N^{\text{adj}} \sim N^{-1/2}$, much larger than $1 - p_N \sim (\log N)/N$. Away from the ends the single R of bin 1 needs 2 switches, so $S_{1,N-1} \approx Nu^2$ reaches 1 only at $u \approx N^{-1/2}$ (at $p = 0, 1$ there are also ties between bins of probability 0).

4.1 A Bessel estimate uniform in the bin

Bin a has about $(a^r/r!)(b^r/r!)$ words with r runs of each letter, each of weight about u^{2r} . The sum depends only on abu^2 , so a bin near the edge acts like the middle of a shorter row with $\sqrt{ab}$ in place of $N/2$. Put

$$s = \sqrt{ab}, \quad A = \frac{ab}{a+b}, \quad c = \frac{a+b}{2\sqrt{ab}}, \quad \eta = \frac{1}{c}, \quad x = su, \quad J_c(x) = I_0(2x) + cI_1(2x),$$

and $\widehat{S}_{a,b}(u) = 2uJ_c(su)$.

Theorem 4.3 (uniform estimate for every bin). *For all positive integers a, b and all $u > 0$,*

$$0 \leq 1 - \frac{S_{a,b}(u)}{\widehat{S}_{a,b}(u)} \leq \frac{(a+b)u^2}{2} + u = \frac{x^2}{2A} + \frac{x}{s}. \quad (14)$$

In particular, for fixed N and u the bound does not depend on the bin. The inequality still holds when the right side exceeds 1, but then it gives no lower bound on $S_{a,b}$.

Proof. Let $U_r = U_r(a)$ and $V_r = U_r(b)$ as in Lemma 3.2(b), and $H = 1/a + 1/b = 1/A$. Lemma 3.2(a), applied to the $2r$ clipped factors of $U_r V_r$, gives

$$0 \leq 1 - U_r V_r \leq \frac{H}{2} r(r+1). \quad (15)$$

The same argument for $E_r = (aU_{r+1}V_r + bU_rV_{r+1})/(a+b)$ gives

$$\begin{aligned} 0 \leq 1 - E_r &\leq \frac{(r+1)(r+2)}{a+b} + \frac{r(r+1)}{2(a+b)} \left(\frac{a}{b} + \frac{b}{a} \right) \\ &= \frac{H}{2} r(r+1) + \frac{2(r+1)}{a+b}. \end{aligned} \quad (16)$$

Since $\binom{a-1}{r} = a^r U_r / r!$, (13) becomes

$$S_{a,b}(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} U_r V_r + c \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} E_r \right], \quad (17)$$

and dropping $U_r V_r$ and E_r gives $\widehat{S}_{a,b}(u)$. So by Lemma 3.2(c), with $I_j = I_j(2x)$, $(\widehat{S}_{a,b} - S_{a,b})/(2u)$ lies between 0 and

$$\frac{H}{2} (x^2 I_0 + x I_1 + c x^2 I_1) + \frac{2c x I_0}{a+b} = \left(\frac{x^2}{2A} + \frac{x}{s} \right) J_c(x),$$

since $H/2 = c/s = 1/(2A)$ and $2c/(a+b) = 1/s$. $\square$

Let $x_{a,b} = su_{a,b}$ and let x_B be the positive root of

$$G_\eta(x_B) = A, \quad G_\eta(x) = x\{I_1(2x) + \eta I_0(2x)\}. \quad (18)$$

Differentiating term by term gives

$$G'_\eta(x) - 2G_\eta(x) = 2x(1-\eta)\{I_0(2x) - I_1(2x)\} + \eta I_0(2x) \geq 0, \quad (19)$$

since $I_0(2x) - I_1(2x) = \frac{1}{\pi} \int_0^\pi e^{2x \cos \theta} (1 - \cos \theta) d\theta > 0$ by [28, Eq. 10.32.3]. So $(\log G_\eta)' \geq 2$ for $x > 0$ and $0 \leq \eta \leq 1$.

Corollary 4.4 (quantitative crossing bound). *We have $x_{a,b} \geq x_B$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \leq x_{a,b} - x_B \leq -\frac{1}{2} \log(1 - \delta_{a,b}). \quad (20)$$

Proof. Since $c\eta = 1$ and $2c/s = 1/A$, we have $\widehat{S}_{a,b}(u) = G_\eta(su)/A$. At the crossing $S_{a,b} = 1$, so Theorem 4.3 gives $A \leq G_\eta(x_{a,b}) \leq A/(1 - \delta_{a,b})$. As $G_\eta(x_B) = A$, this gives $x_{a,b} \geq x_B$, and integrating $(\log G_\eta)' \geq 2$ from x_B to $x_{a,b}$ gives (20). For $\eta = 1$, where $2G_1 = F$, this is the proof of Theorem 3.3(ii). $\square$

Remark 4.5 (Jacobi polynomials). The Bessel forms in this paper come from the confluence of Jacobi, Laguerre and Bessel functions near the point 1. For $a \leq b$ and $0 \leq t < 1$,

$$\sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r} t^r = {}_2F_1(1-a, 1-b; 1; t) = (1-t)^{a-1} P_{a-1}^{(0, b-a)}\left(\frac{1+t}{1-t}\right),$$

and for $b > a$ and $0 \leq u < 1$,

$$\sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r+1} u^{2r+2} = \frac{(b-1)u^2}{a} (1-u^2)^{a-1} P_{a-1}^{(1, b-a-1)}\left(\frac{1+u^2}{1-u^2}\right).$$

Near 1, Jacobi polynomials of large degree behave like Bessel functions (the Mehler–Heine formula [28, Eq. 18.11.5]), and Elliott [13] gives expansions in modified Bessel functions uniform near 1. For fixed a and $y = bu^2$, as $b \rightarrow \infty$, [28, Eq. 18.7.21] turns the last sum into $(y/a)L_{a-1}^{(1)}(-y) = H_a(y)$ of (25), and the other terms tend to 0. Temme [27] gives asymptotic estimates of Gegenbauer, Laguerre and Jacobi polynomials. These forms are classical. But Elliott’s expansions have fixed parameters and no error bounds, while here $b-a$ can be as large as $N-2$, and Theorem 4.3 gives an explicit relative error at every N .

Lemma 4.6 (inverting $z + \frac{1}{2} \log z$). *Fix $K > 0$. Let $z \geq 1$ and $\ell \geq 2$, and suppose that $z + \frac{1}{2} \log z = \ell + C + B/z + \omega$ with $|C|, |B| \leq K$ and $|\omega| \leq 1$. Then*

$$z = \ell - \frac{1}{2} \log \ell + C + \frac{\frac{1}{4} \log \ell - \frac{1}{2} C + B}{\ell} + O_K\left(\frac{(\log \ell)^2}{\ell^2} + |\omega|\right).$$

Proof. Put $w = z - \ell$. As $z \geq 1$, $\log z \geq 0$ and $|B/z| \leq K$, so $z \leq \ell + 2K + 1$ and $|w| \leq 2K + 1 + \frac{1}{2} \log(\ell + 2K + 1) = O_K(\log \ell)$, proving the claim for ℓ bounded in terms of K . For larger ℓ , $|w| \leq \ell/2$, so $\log z = \log \ell + w/\ell + O((w/\ell)^2)$ and $B/z = B/\ell + O_K(|w|/\ell^2)$, and $w = C - \frac{1}{2} \log \ell - w/(2\ell) + B/\ell + \omega + O_K((\log \ell)^2/\ell^2)$. This gives first $w = d + O_K(\log \ell/\ell + |\omega|)$ with $d = C - \frac{1}{2} \log \ell$, and then $w = d + (B - d/2)/\ell + O_K((\log \ell)^2/\ell^2 + |\omega|)$. $\square$

4.2 Every growing distance from an edge

Theorem 4.7 (uniform crossing expansion). *Suppose $1 \leq a \leq b$ and $a \rightarrow \infty$, with no restriction on the relative growth of a and b . Put*

$$\ell = \log A, \quad C(\eta) = \frac{1}{2} \log(8\pi) - \log(1 + \eta), \quad B(\eta) = \frac{3 - \eta}{8(1 + \eta)}.$$

Uniformly over $b \geq a$,

$$2\sqrt{ab}u_{a,b} = \ell - \frac{1}{2}\log \ell + C(\eta) + \frac{\frac{1}{4}\log \ell - \frac{1}{2}C(\eta) + B(\eta)}{\ell} + O\left(\frac{(\log \ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \quad (21)$$

The same expansion and error order hold with $1 - p_{a,b}$ in place of $u_{a,b}$.

Proof. Here $a/2 \leq A \leq a$, $s \geq a$ and $0 < \eta \leq 1$. At $x = \log a$ the bound of Theorem 4.3 is at most $2(\log a)^2/a$, while $G_\eta(x)/A \geq xI_1(2x)/a \rightarrow \infty$. So $S_{a,b} > 1$ there, $x_{a,b} \leq \log a$ for large a , uniformly in b , and Corollary 4.4 gives

$$x_{a,b} - x_B = O((\log a)^2/a). \quad (22)$$

Write $z = 2x_B$, so $z \rightarrow \infty$ as $A \rightarrow \infty$. By [28, Eq. 10.40.1], $I_1(z) + \eta I_0(z) = (1 + \eta)e^z(2\pi z)^{-1/2}(1 - B(\eta)/z + O(z^{-2}))$ uniformly in $0 \leq \eta \leq 1$, so (18) becomes $z + \frac{1}{2}\log z = \ell + C(\eta) + B(\eta)/z + O(z^{-2})$. As $|C(\eta)| \leq 2$ and $0 < B(\eta) \leq \frac{3}{8}$, Lemma 4.6 expands z , and (22) moves this to $2x_{a,b} = 2\sqrt{ab}u_{a,b}$. Finally $2s(u - u/(1 + u)) \leq 2x^2/s = O((\log a)^2/a)$ at $u = u_{a,b}$. $\square$

Hence, uniformly in b ,

$$1 - p_{a,b} \sim \frac{\log a}{2\sqrt{ab}} \quad (a \rightarrow \infty, b \geq a). \quad (23)$$

If $a = o(b)$, then $\eta \rightarrow 0$ and $\log A = \log a + o(1)$, so

$$2\sqrt{ab}(1 - p_{a,b}) = \log a - \frac{1}{2}\log \log a + \frac{1}{2}\log(8\pi) + o(1). \quad (24)$$

If $a/N \rightarrow \alpha \in (0, \frac{1}{2}]$ at any rate, then

$$2\sqrt{a(N-a)}(1 - p_{a,N-a}) = \log N - \frac{1}{2}\log \log N + \log\{\alpha(1 - \alpha)\} + C(2\sqrt{\alpha(1 - \alpha)}) + o(1).$$

At the center, $a = b = N/2$ gives $A = N/4$ and $\eta = 1$, and changing from $\log A$ to $\log N$ recovers (11).

4.3 Fixed bins and the Laguerre edge law

If a is fixed and $b \rightarrow \infty$, each run of R 's away from the ends needs 2 switches, giving u^2 , and can go in about b places. So we use $y = bu^2$. Define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \quad (25)$$

with the standard Laguerre polynomial $L_{a-1}^{(1)}$. Let y_a be the positive root of $H_a(y_a) = 1$ and $c_a = \sqrt{y_a}$. Then $y_1 = 1$, $y_2 = \sqrt{3} - 1$, and y_a decreases strictly in a , by coefficient comparison.

Theorem 4.8 (fixed-edge expansion). *For each fixed positive integer a , as $b \rightarrow \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \quad (26)$$

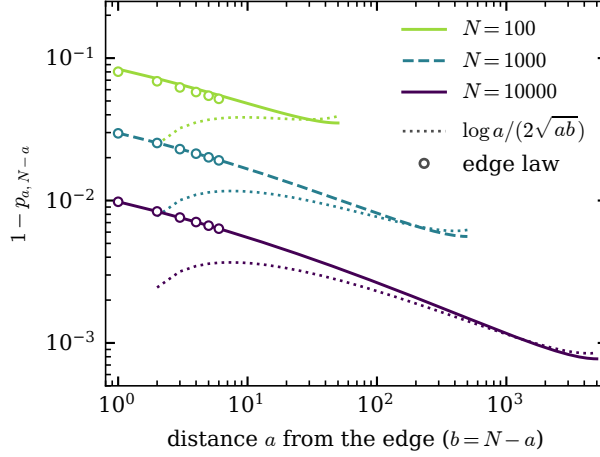


Figure 4: The crossing $1 - p_{a, N-a}$ of each bin against its distance a from the nearer edge, for $N = 100, 1000$ and 10^4 . The dotted lines are the leading law (23), and the circles are the edge law (27) for $a \leq 6$. The right end of each line is the central crossing $1 - p_N$.

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2)H'_a(y_a)/H_a(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 for every fixed bin. For the switching probability,

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1 + c_a^2}{b} + O_a(b^{-3/2}). \quad (27)$$

Proof. Put $\epsilon = b^{-1/2}$, $y = bu^2$ and $V_r = U_r(b)$. For $b \geq a$, (13) reads

$$\begin{aligned} S_{a,b}(u) &= 2\sqrt{y/b} \sum_r \binom{a-1}{r} \frac{y^r}{r!} V_r \\ &\quad + \frac{y}{b} \sum_r \binom{a-1}{r+1} \frac{y^r}{r!} V_r + y \sum_r \binom{a-1}{r} \frac{y^r}{(r+1)!} V_{r+1}. \end{aligned} \quad (28)$$

These sums have at most a terms, and with $V_r = 1$ they are $H'_a(y)$, $H''_a(y)$ and $H_a(y)$. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ and using $\sum_r (r+1)(r+2) \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = y^2 H''_a(y) + 2y H'_a(y)$ gives, uniformly near $y = y_a$,

$$\begin{aligned} S_{a,b}(u) &= H_a(y) + 2\sqrt{y} H'_a(y) \epsilon \\ &\quad + \{(y - y^2/2)H''_a(y) - yH'_a(y)\} \epsilon^2 + O_a(\epsilon^3). \end{aligned} \quad (29)$$

With $b = \epsilon^{-2}$ the right side of (28) is analytic in (ϵ, y) near $(0, y_a)$, and $H'_a(y_a) > 0$. So the implicit-function theorem and the uniqueness of the crossing give $y = y_a + d_1 \epsilon + d_2 \epsilon^2 + O_a(\epsilon^3)$ at $u = u_{a,b}$, and matching coefficients in (29) gives $d_1 = -2c_a$ and $d_2 = 2 + y_a + (y_a + y_a^2/2)H''_a(y_a)/H'_a(y_a)$. Expanding $u = \epsilon\sqrt{y}$ and then $u/(1+u)$ proves both expansions. $\square$

Theorem 4.9 (matching the edge and Bessel regimes). *For every positive integer a and every $x > 0$,*

$$0 \leq 1 - \frac{aH_a(x^2/a)}{xI_1(2x)} \leq \frac{x^2}{2a}. \quad (30)$$

Consequently, with $\ell = \log a$ and $C_0 = \frac{1}{2} \log(8\pi)$,

$$2\sqrt{ay_a} = \ell - \frac{1}{2} \log \ell + C_0 + \frac{\frac{1}{4} \log \ell - \frac{1}{2} C_0 + 3/8}{\ell} + O\left(\frac{(\log \ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \quad (31)$$

Proof. With $U_r = U_r(a)$, $aH_a(x^2/a) = x \sum_{r \geq 0} U_r x^{2r+1}/(r!(r+1)!)$, so Lemma 3.2(b) and the second identity of Lemma 3.2(c) give (30). Put $x_a = \sqrt{ay_a}$, so $aH_a(x_a^2/a) = a$, and $x_a < \log a$ for large a by (30). So $a \leq G_0(x_a) \leq a/(1 - x_a^2/(2a))$, and as in Corollary 4.4, $x_a - x_0 = O((\log a)^2/a)$, where $G_0(x_0) = a$. So $z = 2x_0$ satisfies $z + \frac{1}{2} \log z = \ell + C_0 + 3/(8z) + O(z^{-2})$ by [28, Eq. 10.40.1], as in the proof of Theorem 4.7 with $\eta = 0$, $A = a$, and Lemma 4.6 gives the expansion. $\square$

Figure 4 compares these laws with the exact crossings.

4.4 The moving boundary layer

The all-bin expansion also gives the shape of a row at odds $u = v_N > 0$. Let

$$\mathcal{L}_N = \#\{1 \leq k \leq \lfloor N/2 \rfloor : P_N(k) < P_N(0)\}.$$

Corollary 4.10 (boundary-layer width). *Suppose*

$$\Omega_N = \frac{1}{\sqrt{N} v_N} \longrightarrow \infty, \quad \frac{\log \Omega_N}{N v_N} \longrightarrow 0.$$

Then

$$\mathcal{L}_N \sim \Omega_N^2 \log^2 \Omega_N = \frac{\log^2(1/(\sqrt{N} v_N))}{N v_N^2}. \quad (32)$$

In particular, if $v_N = N^{-\gamma}$ with $1/2 < \gamma < 1$, then

$$\mathcal{L}_N \sim (\gamma - \frac{1}{2})^2 N^{2\gamma-1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ for all sufficiently large N .

Proof. Set $d_N = \Omega_N^2 \log^2 \Omega_N$. Then $d_N \rightarrow \infty$ and $d_N/N = (\log \Omega_N/(N v_N))^2 \rightarrow 0$. Fix $\varepsilon \in (0, 1)$ and integers $a_{\pm} = (1 \pm \varepsilon)d_N + O(1)$. Then $\log a_{\pm} \sim 2 \log \Omega_N$ and $\sqrt{a_{\pm}(N - a_{\pm})} v_N \sim \sqrt{1 \pm \varepsilon} \log \Omega_N$, so Theorem 4.7 gives $u_{a_{\pm}, N-a_{\pm}}/v_N \rightarrow 1/\sqrt{1 \pm \varepsilon}$. Bin k is below the endpoint exactly when $v_N < u_{k, N-k}$, and $u_{k, N-k}$ decreases strictly for $k \leq N/2$ by Proposition 4.1. So $a_- \leq \mathcal{L}_N < a_+$ for large N , and we let $\varepsilon \downarrow 0$. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint. $\square$

5 Boundary conditions and a spatial crossing window

The periodic chain adds the bond between the last and the first spin to the Ising chain of the introduction. So its weight is proportional to $\exp(J \sum_{i=1}^N \sigma_i \sigma_{i+1})$ with $\sigma_{N+1} = \sigma_1$, and again $u = e^{-2J}$. Antal, Droz and Rácz [5] give the magnetization laws of both chains at a fixed scale, and Dantchev and Rudnick [11] give exact free and periodic coefficients. This chain is not the free walk on the cylinder of Remark 2.7.

Write $R_{N,k}^F = S_{k,N-k}$ and $R_{N,k}^P$ for the bin-to-endpoint ratios under free and periodic boundaries. Put $a = k$, $b = N - k$ and $s = \sqrt{ab}$. The classical cyclic count is

$$R_{N,k}^P(u) = \sum_{r=1}^{\min(a,b)} \frac{N}{r} \binom{a-1}{r-1} \binom{b-1}{r-1} u^{2r}. \quad (33)$$

To see this, choose compositions of a and of b into r positive runs and a position for the start of a distinguished run of R 's, and divide by the r choices of that run. The normalization of the periodic measure cancels in the ratio. In particular $R_{N,1}^P = Nu^2$, which gives the adjacent crossing $J = (\log N)/4$ of Stepanyan et al. [26]. Every interior ratio increases strictly from 0 to ∞ in $u > 0$, and the ratios increase strictly toward the middle, apart from the symmetric middle tie, by the coefficient comparison of Proposition 4.1.

Theorem 5.1 (periodic relative error). *For $N \geq 2$, $1 \leq k \leq N - 1$, and $u > 0$, define*

$$\hat{R}_{N,k}^P(u) = \frac{Nu}{\sqrt{k(N-k)}} I_1\left(2u\sqrt{k(N-k)}\right).$$

Then

$$0 \leq 1 - \frac{R_{N,k}^P(u)}{\hat{R}_{N,k}^P(u)} \leq \frac{Nu^2}{2}. \quad (34)$$

Proof. Put $x = su$, $A = ab/N$, $U_r = U_r(a)$ and $V_r = U_r(b)$. Shifting the index in (33) gives $R_{N,k}^P(u) = (Nu/s) \sum_{r \geq 0} U_r V_r x^{2r+1}/(r!(r+1)!)$, so the ratio to $\hat{R}_{N,k}^P$ is an average of the $U_r V_r$ with weights $x^{2r+1}/(r!(r+1)!)$, whose sum is $I_1(2x)$. By (15), $0 \leq 1 - U_r V_r \leq r(r+1)/(2A)$, and the second identity of Lemma 3.2(c) bounds the deficit by $x^2/(2A) = Nu^2/2$. $\square$

We can guess the $\log 2$ in the next theorem. A cyclic word switches an even number of times, and the free words that start and end on different letters give the I_0 term of the free approximation. As $I_0 \approx I_1$ for large arguments, the free ratio is about twice the periodic one near the crossing. Both grow like e^{Nu} , so the periodic crossing needs Nu about $\log 2$ larger. The factor 2 also appears at a fixed scale $T = Nq$, where Antal, Droz and Rácz [5, Eqs. (37) and (47)] find the endpoint atoms $1/(2 \cosh T)$ (periodic) and $e^{-T}/2$ (free), whose ratio tends to 2 while the densities at zero magnetization agree to leading order.

Theorem 5.2 (periodic crossing and boundary shift). *Let u_N^P be the positive root of $R_{N,m_N}^P(u) = 1$, and let $u_N^F = u_N$ be the free central root. Put $L = \log N$ and $c_P = \frac{1}{2} \log(\pi/2)$. Then*

$$Nu_N^P = L - \frac{1}{2} \log L + c_P + \frac{\frac{1}{4} \log L - \frac{1}{2} c_P + \frac{3}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right), \quad (35)$$

$$N(u_N^P - u_N^F) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2} \log 2}{\log N} + o(1/\log N). \quad (36)$$

Replacing Nu_N^P by $N(1 - p_N^P)$ preserves (35). At the periodic crossing the coupling J_N^P satisfies

$$J_N^P = \frac{1}{2}(\log N - \log \log N) + \frac{\frac{1}{4} \log \log N - \frac{1}{2} c_P}{\log N} + O\left(\frac{(\log \log N)^2}{(\log N)^2}\right). \quad (37)$$

Proof. At the middle $2s = N + O(1/N)$. At $u = (1 \pm \epsilon)L/N$ with fixed $\epsilon \in (0, 1)$ the bound of Theorem 5.1 tends to 0, and the large-argument form of I_1 makes $\hat{R}^P$ tend to ∞ with the plus sign and to 0 with the minus sign. By monotonicity $Nu_N^P \sim L$. Write $z = Nu_N^P$. At the root $\log \hat{R}^P = O(L^2/N)$ by (34), replacing the argument $2su$ by z changes it by $O(L/N^2)$, and $I_1(z) = e^z(2\pi z)^{-1/2}(1 - 3/(8z) + O(z^{-2}))$ [28, Eq. 10.40.1]. So

$$z + \frac{1}{2} \log z = L + c_P + \frac{3}{8z} + O(z^{-2} + L^2/N),$$

and Lemma 4.6 with $\ell = L$, $C = c_P$ and $B = \frac{3}{8}$ gives (35). As $N(u - u/(1+u)) = O(L^2/N)$ for $Nu = O(L)$, it also holds for $N(1 - p_N^P)$, and $Nu_N = z_N + O(L^2/N)$ by (7). Subtracting (11), where $c_0 = c_P - \log 2$ and the correction is $\frac{1}{8}$, gives (36). Finally $J = -\frac{1}{2} \log u = \frac{1}{2}(L - \log z)$, and $\log z = \log L - (\frac{1}{2} \log L - c_P)/L + O((\log L)^2/L^2)$ gives (37). $\square$

Stepanyan et al. [26, Eq. (16)] approximate the periodic crossing by $\beta J \approx (2 \log N - 1)/5$, and their Fig. 1(b) fits the line $\beta J \approx 0.3864 \log N - 0.21055$. At $N = 10^5$ the exact periodic crossing is $J = 4.576$ (their βJ). Their Eq. (16) gives 4.405, their line gives 4.238, and (37) without its error term gives 4.578. The line is accurate for $10 \leq N \leq 100$ (at $N = 100$ it gives 1.569 and the exact value is 1.574), but it has no $\log \log N$ term.

The factor e^{-2y^2} is easy to guess. Moving $yN/\sqrt{\log N}$ away from the middle multiplies the Bessel argument by $2\sqrt{k(N-k)}/N \approx 1 - 2y^2/\log N$. The argument is about $\log N$, so it drops by about $2y^2$, and the ratio, which grows like e to the argument, drops by about the factor e^{-2y^2} .

Theorem 5.3 (spatial crossing window). *Let $X \in \{F, P\}$ and let u_N^X be its exact central crossing. For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \quad k_N(y) = \text{nint}\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, uniformly on compact sets of (t, y) ,

$$R_{N, k_N(y)}^X(u_N^X + t/N) \longrightarrow e^{t-2y^2}. \quad (38)$$

If $M_N^X(t)$ is the number of interior bins with probability strictly above the endpoint probability at that parameter, then

$$\frac{\sqrt{\log N}}{N} M_N^X(t) \longrightarrow \sqrt{2 \max(t, 0)}. \quad (39)$$

Proof. Put $\delta = k/N - \frac{1}{2}$ and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1 - 4\delta^2}$. For the stated rounding and bounded y ,

$$\eta = 1 - \frac{2y^2}{L} + O\left(L^{-2} + \frac{1}{N\sqrt{L}}\right), \quad Nu_N^X = L + O(\log L).$$

So the Bessel argument $Nu\eta$ at (u, k) is $Nu_N^X + t - 2y^2 + o(1)$, uniformly on the compact set. By Theorems 4.3 and 5.1 both ratios are within relative error $O(L^2/N)$ of their Bessel forms, and $I_0(z)$ and $I_1(z)$ equal $e^z(2\pi z)^{-1/2}(1 + O(1/z))$ [28, Eq. 10.40.1], so the prefactors relative to their central values tend to 1 uniformly. The change of argument gives e^{t-2y^2} , and the exact central ratio is 1, which proves (38).

The interior ratios increase toward the middle. If $t \leq 0$, the central ratio is at most 1, so no interior bin is above the endpoint. If $t > 0$ and $0 < \epsilon < \sqrt{t/2}$, then for large N the bins at scaled distance less than $\sqrt{t/2} - \epsilon$ are above the endpoint and those at distance $\sqrt{t/2} + \epsilon$ are below, and by monotonicity so are all farther bins. Count the bins in between, divide by $N/\sqrt{L}$ and let $\epsilon \downarrow 0$. Rounding changes the count by $O(1)$. $\square$

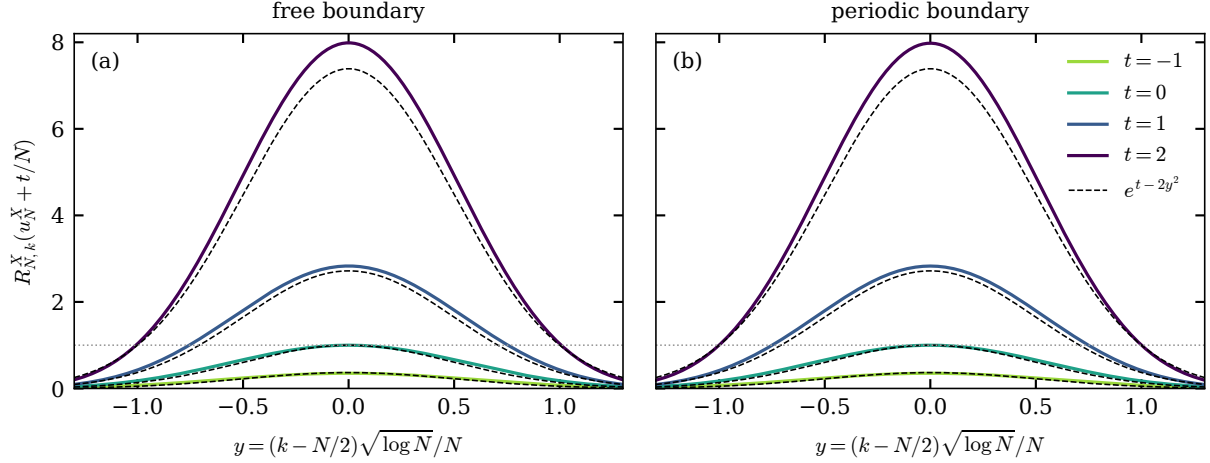


Figure 5: The exact ratios $R_{N,k}^X(u_N^X + t/N)$ at $N = 10^6$ for free (a) and periodic (b) boundaries, with the limit e^{t-2y^2} of Theorem 5.3 (dashed). Here $y = (k - N/2)\sqrt{\log N}/N$, and the bins above the dotted line are above the endpoint height. The convergence is slow, and at $t = 2$ and $y = 0$ the relative error is still 0.08.

The window has width about $1/N$ in p and about $N/\sqrt{\log N}$ in space. A weak limit theorem would not give this, since it says nothing about a single bin such as the endpoint. Figure 5 shows the exact ratios at $N = 10^6$.

6 A uniform local law

So far we compared each bin with an endpoint. Now we approximate $P_N(k)$ itself by a single formula, uniformly in $q \leq \frac{1}{2}$, from the diffusive regime (q fixed) to the ballistic regime ($T = Nq$ bounded). Throughout, $n_1 = k - 1$, $n_2 = N - k - 1$ and $n_{\min} = \min(n_1, n_2)$.

Theorem 6.1 (a difference of 2 binomials). *Let $N \geq 2$, $1 \leq k \leq N - 1$ and $0 < q < 1$. Let $Y_1 \sim \text{Bin}(n_1, q)$ and $Y_2 \sim \text{Bin}(n_2, q)$ be independent, and put $D = Y_1 - Y_2$. Then*

$$P_N(k) = \frac{q}{2} [2\mathbb{P}(D = 0) + \mathbb{P}(D = 1) + \mathbb{P}(D = -1)]. \quad (40)$$

Proof. We rewrite Theorem 2.1, where $a - 1 = n_1$ and $b - 1 = n_2$. Note that $\mathbb{P}(Y_1 = r)\mathbb{P}(Y_2 = r') = \binom{n_1}{r} \binom{n_2}{r'} q^{r+r'} p^{N-2-r-r'}$. The term $j = 2r + 1$ of Theorem 2.1 is $\binom{n_1}{r} \binom{n_2}{r} p^{N-2-2r} q^{2r+1} = q\mathbb{P}(Y_1 = r)\mathbb{P}(Y_2 = r)$, and summing over r gives $q\mathbb{P}(D = 0)$. The term $j = 2r + 2$ is

$$\frac{1}{2} \left[\binom{n_1}{r+1} \binom{n_2}{r} + \binom{n_1}{r} \binom{n_2}{r+1} \right] p^{N-3-2r} q^{2r+2} = \frac{q}{2} [\mathbb{P}(Y_1 = r+1)\mathbb{P}(Y_2 = r) + \mathbb{P}(Y_1 = r)\mathbb{P}(Y_2 = r+1)],$$

and summing over r gives $\frac{q}{2} [\mathbb{P}(D = 1) + \mathbb{P}(D = -1)]$. $\square$

For $k = 1$ we have $n_1 = 0$ and $D = -Y_2$, so $\mathbb{P}(D = 0) = p^{N-2}$, $\mathbb{P}(D = -1) = (N-2)qp^{N-3}$ and $\mathbb{P}(D = 1) = 0$. By (40) and symmetry,

$$P_N(1) = P_N(N-1) = qp^{N-2} + \frac{1}{2}(N-2)q^2p^{N-3}. \quad (41)$$

With $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, the 4 bins $0, 1, N-1, N$ are exact, and from now on $N \geq 4$ and $2 \leq k \leq N-2$.

Next we tilt D to mean 0. For $s > 0$ put

$$r_1 = \frac{qs}{p+qs}, \quad r_2 = \frac{q}{ps+q}, \quad v_i = r_i(1-r_i), \quad \Lambda(s) = (p+qs)^{n_1}(p+q/s)^{n_2}.$$

Under the tilted law $\mathbb{P}_s$, $Y_1 \sim \text{Bin}(n_1, r_1)$ and $Y_2 \sim \text{Bin}(n_2, r_2)$ are independent, and $\mathbb{P}(D = j) = \Lambda(s)s^{-j}\mathbb{P}_s(D = j)$ (Lemma B.1). Since $n_1, n_2 \geq 1$, there is a unique $s > 0$ with $n_1r_1 = n_2r_2 =: m$, so that $\mathbb{E}_s D = 0$ (Lemma B.2). We fix this s , let $x = n_1v_1 + n_2v_2$ be the variance of D under $\mathbb{P}_s$, and define

$$U_N(k) = \frac{q}{2}\Lambda(s)e^{-x}[2I_0(x) + (s+s^{-1})I_1(x)]. \quad (42)$$

At the center of an even row, $n_1 = n_2$ gives $s = 1$, $r_1 = r_2 = q$ and $\Lambda = 1$, so

$$U_N(N/2) = qe^{-x}(I_0(x) + I_1(x)), \quad x = (N-2)pq. \quad (43)$$

Note that by (40) and the tilt, $P_N(k) = \frac{q}{2}\Lambda(s)[2\mathbb{P}_s(D=0) + s^{-1}\mathbb{P}_s(D=1) + s\mathbb{P}_s(D=-1)]$. The law $e^{-x}I_{|j|}(x)$ on $\mathbb{Z}$ is the symmetric Skellam law, the law of a difference of 2 independent Poisson variables with mean $x/2$, and it has mean 0 and variance x . So $U_N(k)$ replaces the tilted D by the Skellam law with the same mean and variance. This law is exact for a Poisson difference, which D resembles when q is small, and close to Gaussian when x is large. At the center D is a sum of $N/2 - 1$ independent variables with $\mathbb{P}(\pm 1) = pq$, and replacing it by this Skellam law is the approximation that Čekanavičius and Liaudanskaitė [10, Eq. (3)] attribute to Presman. The tilt extends it off the center.

Let $\kappa = 1 - \cos t$ and write $\langle h \rangle$ for the average of $h(t)$ against the density proportional to $e^{x \cos t}$ on $[-\pi, \pi]$. With $\mathcal{R} = \mathcal{R}(x) = I_1(x)/I_0(x)$, Lemma B.6 gives

$$\langle \kappa^2 \rangle = 2 - 2\mathcal{R} - \frac{\mathcal{R}}{x}, \quad \langle \kappa^3 \rangle = 4 - 4\mathcal{R} - \frac{3\mathcal{R}}{x} - \frac{2\mathcal{R}}{x^2} + \frac{1}{x}.$$

Put $V = n_1v_1^2 + n_2v_2^2$ and $M = m \max(r_1, r_2)$.

Theorem 6.2 (uniform local law). *Let $N \geq 4$, $2 \leq k \leq N-2$ and $0 < q < 1$. Then $P_N(k) = U_N(k)(1 + \theta)$ with*

$$|\theta| \leq \mathcal{E} := \frac{1}{\mathcal{R}(x)} \left[\left(4V + \frac{64}{3}M \right) \langle \kappa^2 \rangle + \frac{1024}{9}M^2 \langle \kappa^3 \rangle \right].$$

We sketch the proof. By Fourier inversion, $P_N(k)$ and $U_N(k)$ are $\frac{q}{2}\Lambda(s)\frac{1}{2\pi}\int_{-\pi}^{\pi} f\omega dt$ with $\omega(t) = 2 + se^{it} + s^{-1}e^{-it}$, where f is $\chi(t) = \mathbb{E}_s e^{itD}$ for $P_N(k)$ and the Skellam characteristic function $g(t) = e^{x(\cos t - 1)}$ for $U_N(k)$. Since $|1 - r + re^{it}|^2 = 1 - 2v\kappa = e^{-2v\kappa}(1 + O(v^2\kappa^2))$ with $v = r(1-r)$, $|\chi| \leq g$ and $g - |\chi| = O(V\kappa^2)g$. The phase of χ has no linear term since $\mathbb{E}_s D = 0$, so it is $O(M|t|^3)$. Hence $|\theta|$ is at most a constant times $((V+M)\langle \kappa^2 \rangle + M^2\langle \kappa^3 \rangle)/\mathcal{R}(x)$, where $\langle \kappa^j \rangle = O(\min(1, x^{-j}))$ and $1/\mathcal{R}(x) \leq 1 + 2/x$. The proof with explicit constants is in Appendix B.

Corollary 6.3. *Let $N \geq 4$, $2 \leq k \leq N-2$ and $0 < q \leq \frac{1}{2}$, and put $r_{\max} = \max(r_1, r_2)$. Then*

$$|\theta| \leq \min \left\{ \frac{22334}{n_{\min}}, 221r_{\max} + 5825r_{\max}^2 \right\}.$$

Also $r_{\max} \leq \delta_T := 2T/\sqrt{N} + 4T^2/N$, so $|\theta| \leq 221\delta_T + 5825\delta_T^2$ for every bin $2 \leq k \leq N-2$. At the center of an even row, $r_{\max} = q$ and $n_{\min} = N/2 - 1$.

So the relative error tends to 0 uniformly in $0 < q \leq \frac{1}{2}$ as $n_{\min} \rightarrow \infty$, and uniformly over $2 \leq k \leq N - 2$ as $T/\sqrt{N} \rightarrow 0$, which covers the ballistic regime $T = O(1)$ up to the edges. The constants are crude. The largest $n_{\min}|\theta|$ we observed is 0.686, at $N = 10^4$ and $q = \frac{1}{2}$. It is the largest over $2 \leq k \leq 400$ at $N = 10^4$ with $q = \frac{1}{2}$ or 0.3, and over every bin of our grid with $N \leq 1000$ and $10^{-5} \leq q \leq \frac{1}{2}$. The first bound has the constant 22334 instead, so it exceeds 1 unless $n_{\min} > 22334$. Near an edge the largest error we observed is $|\theta| = 0.1617$ ($k = 3$, $N = 10^4$, $q = \frac{1}{2}$), an observation and not a bound. For $q > \frac{1}{2}$ Theorem 6.2 still holds, but $\mathcal{E}$ need not be small, and a uniform form is open. Near $q = 1$ the symmetric kernel fails once $n_1 \neq n_2$. We found $\theta = 0.578$ at the middle bin $k = 50$ of the odd row $N = 101$ with $q = 0.999$, while $\theta \approx 0$ at $N = 100$, $k = 50$.

Replacing $e^{-x}I_j(x)$ in (42) by $(2\pi x)^{-1/2}$ gives the Gaussian saddle point $G_N(k) = \frac{q}{2}\Lambda(s)(2 + s + s^{-1})(2\pi x)^{-1/2}$. We also define the transfer-matrix form $\text{TM}_N = 2(2\pi(N-1)p/q)^{-1/2}$, the Gaussian local form for $\mathbb{P}(X_N = 0)$ with the variance $(N-1)p/q$ that λ_+^{N-1} gives in the proof of Theorem 2.5. The factor 2 is the span of the lattice of $X_N = 2k - N$.

Proposition 6.4 (Gaussian saddle points). *Fix an even $N \geq 4$. As $q \rightarrow 0$,*

$$\frac{P_N(N/2)}{q} \rightarrow 1, \quad \frac{G_N(N/2)}{P_N(N/2)} \sim \frac{2}{\sqrt{2\pi(N-2)q}} \rightarrow \infty, \quad \frac{\text{TM}_N}{P_N(N/2)} \sim \frac{2}{\sqrt{2\pi(N-1)q}} \rightarrow \infty.$$

Proof. The term $j = 1$ of Theorem 2.1 is qp^{N-2} and the other terms are $O(q^2)$, so $P_N(N/2) = q + O(q^2)$. At the center $s = 1$, $\Lambda = 1$ and $x = (N-2)pq$, so $G_N(N/2) = 2q/\sqrt{2\pi(N-2)pq}$. Also $\text{TM}_N = 2q/\sqrt{2\pi(N-1)pq}$. Let $q \rightarrow 0$. $\square$

More generally, since $P_N(N/2)/q \rightarrow 1$, any approximation $A_N(q)$ of the center with $A_N(q)/q \rightarrow \infty$ has relative error $P_N(N/2)/A_N(q) - 1 \rightarrow -1$. Both saddle points are of this kind. We make no claim about other Gaussian forms. The Bessel form avoids this because $e^{-x}I_j(x)$ is exact for the Poisson difference that D approaches in the ballistic regime.

Remark 6.5 (the center crossing). For even $N \geq 4$ let $\mathcal{G}(q)$ be the logarithm of the ratio of $U_N(N/2)$ to $P_N(0) = \frac{1}{2}p^{N-1}$. By $I'_0 = I_1$, $I'_1 = I_0 - I_1/x$ [28, Eqs. 10.29.2, 10.29.3] and $I_1 \leq I_0$, we get $\mathcal{G}'(q) \geq \frac{1}{2pq} + \frac{N-1}{p} > N-1$ on $(0, \frac{1}{2}]$. Also $\mathcal{G} \rightarrow -\infty$ as $q \rightarrow 0$ and $\mathcal{G}(\frac{1}{2}) \geq (N-1)\log 2 - \frac{N-2}{4} > 0$, so $\mathcal{G}$ has a unique root q_U there, an approximate crossing. In Table 3, which is purely numerical, the relative error of q_U is 16 to 36 times smaller than that of the Bessel root u_B of Theorem 3.3. Put $\vartheta_N = 221q_N + 5825q_N^2$. If $\vartheta_N < 1$, Corollary 6.3 gives $|\mathcal{G}(q_N)| \leq \vartheta_N/(1 - \vartheta_N)$, and the mean value theorem gives $|q_U - q_N| \leq \vartheta_N/((1 - \vartheta_N)(N-1))$. By our computation $\vartheta_N \geq 1$ for every even $N < 1454$, and beyond it this bound is weaker than Theorem 3.3(ii) at every N we computed.

N	p_N	$q_U/q_N - 1$	$u_B/u_N - 1$	ratio
100	0.9649663263	-1.477×10^{-3}	-2.39×10^{-2}	0.062
400	0.9881246429	-3.083×10^{-4}	-7.61×10^{-3}	0.041
1600	0.9962346137	-6.469×10^{-5}	-2.31×10^{-3}	0.028

Table 3: The root q_U of $U_N(N/2) = P_N(0)$ against the Bessel root u_B of Theorem 3.3. The ratio is $(q_U/q_N - 1)/(u_B/u_N - 1)$. Numerical values, not bounds.

At the center our kernel is the Skellam law of [10, Eq. (3)], as noted above. Čekanavičius and Liaudanskaitė [10] and Čekanavičius and Vellaisamy [9] give total-variation, local and Wasserstein bounds with unspecified absolute constants for chains outside our regime. One is a symmetric

3-state chain whose step is 0 with probability at least $\frac{14}{15}$. The other is a 2-state chain with $\mathbb{P}(1 \mid 1) \leq \frac{1}{2}$, and some of its bounds also need $\mathbb{P}(1 \mid 0) \leq \frac{1}{30}$. Neither allows both states to persist, as our $p \rightarrow 1$ does. Jensen [18] conditions the count S_n on $0 < S_n < n$ and patches small tilted variances, and his relative error stays bounded (about 0.18, found partly by numerics) but does not tend to 0. In the literature we searched we did not find a tilted Skellam kernel off the center with a relative local error, explicit constants and uniformity in $q \leq \frac{1}{2}$.

7 Verification

The checks for all 4 papers are described in `SUPPLEMENT.md`, in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>, together with the manuscript source and the recorded results. The scripts are standard-library Python, and `python -B verify_all.py --paper a` runs the ones for this paper (`--full` adds 2 long runs of about 21 minutes). They check the exact law, the lattice-path counts, the orderings, the Bessel bounds and the asymptotic formulas, and where possible each reported number is computed in 2 independent ways. Apart from Remark 3.9, which is computer-assisted throughout, a proof rests on only 2 computations. These are the integer certificates for $N \leq 174$ in Theorem 3.4 and Proposition 3.8, made and rechecked by `verify_every_n.py` and `verify_every_n_guards.py`, and the evaluation at $N = 2774$ in Corollary 3.7(d). Before the main computations (the monotonicity and Bessel-ratio checks, the every- N certificates and the local law) we wrote predictions and conditions that would refute them in `data/ledger.md`, `data/ledger_every_n.md` and `data/ledger_llt.md`, and we recorded the outcomes there. Some small- N predictions for Theorem 3.4 were not blind, as the ledger says. The Lean 4 proofs are in `lean/PaperA`, and `lean/README.md` lists them. They include the exact law, the numerator tables, the variance and the normal limit, the orderings of the crossings, the uniform Bessel bounds and root comparisons, the leading terms $1 - p_N \sim (\log N)/N$ and $2\sqrt{ab}u_{a,b} \sim \log a$, the finite parts of the edge and periodic results, Lemmas 3.5 and 3.6 and Theorem 3.4 for every $N \geq 2$. The finite parts are the polynomial H_a and its root y_a , the exact series (28), the cyclic count (33) and the adjacent periodic crossing. For Theorem 4.8 only the algebra of the coefficients is in Lean. For $N \leq 174$ the Lean kernel checks the same certificates in exact integer arithmetic, with its own rational bounds on ζ_N , so the whole theorem is proved in Lean. `lean/build-log.txt` shows that every theorem uses only the standard axioms. Remarks 2.6, 2.7, 4.5 and 3.9, Corollary 3.7 beyond part (a), Proposition 3.8, Lemma 4.6, Theorem 4.7 beyond its leading term, the expansions of Theorem 4.8, the expansion (31) of Theorem 4.9, Theorem 5.2, even its leading term, Corollary 4.10, Theorem 5.3 and Section 6 are not in Lean.

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A The crossing at every N

In this appendix the letters $U, W, W', r, r', t, t', \tau, \kappa, \ell, m, B, D, \beta, \Delta$ and δ have the local meanings given below.

A.1 Proof of Lemma 3.6

For integer n , $I_n(z) = \frac{1}{\pi} \int_0^\pi e^{z \cos t} \cos(nt) dt$ [28, Eq. 10.32.3], so $I_0(z) + I_1(z) = \frac{1}{\pi} \int_0^\pi e^{z \cos t} (1 + \cos t) dt$. Substitute $s = 1 - \cos t$, so that $\sin t = \sqrt{s(2-s)}$ and $1 + \cos t = 2 - s$, and then $s = v/z$.

This gives

$$E(z) = \frac{1}{\sqrt{\pi}} \int_0^{2z} e^{-v} v^{-1/2} \sqrt{1 - \frac{v}{2z}} dv. \quad (44)$$

On $[0, 1]$ we have $\sqrt{1-y} = 1 - \sum_{k \geq 1} c_k y^k$ with every $c_k > 0$, $c_1 = \frac{1}{2}$ and $c_2 = \frac{1}{8}$, and $\sum_k c_k = 1$ since the series tends to 0 at $y = 1$. Since $\sum_{k \geq 3} c_k y^k \leq \frac{3}{8} y^3$, for $0 \leq y \leq 1$

$$q(y) := 1 - \frac{y}{2} - \frac{y^2}{8} - \frac{3y^3}{8} \leq \sqrt{1-y} \leq 1 - \frac{y}{2} - \frac{y^2}{8} \leq p(y) := 1 - \frac{y}{2} - \frac{y^2}{8} + \frac{y^3}{8}.$$

The cubic q decreases and $q(1) = 0$, so $q \leq 0$ on $[1, \infty)$. Also $8p(y) = (y-2)^2(y+1) + 2(y-1)^2 + 2 > 0$ for $y \geq -1$. Put $y = v/(2z)$ in (44) and extend the integral to ∞ . The added piece is at most 0 with q and at least 0 with p . Since $\frac{1}{\sqrt{\pi}} \int_0^\infty e^{-v} v^{k-1/2} dv = \Gamma(k + \frac{1}{2})/\Gamma(\frac{1}{2})$ equals $1, \frac{1}{2}, \frac{3}{4}, \frac{15}{8}$ for $k = 0, 1, 2, 3$, the 2 bounds integrate to the 2 sides of the lemma. The bound $E \leq 1$ follows in the same way from $\sqrt{1-y} \leq 1$. Finally $-\frac{3}{128z^2} + \frac{15}{512z^3} \leq 0$ exactly when $z \geq \frac{5}{4}$.

A.2 Proof of Theorem 3.4 for $N \geq 175$

Fix $N \geq 175$ and write $\zeta = \zeta_N$, $U = \zeta + \frac{1}{8\zeta}$, $r = U/N$ and $n(\zeta) = \sqrt{8\zeta/\pi} e^\zeta$. Then $n(\zeta_N) = N$ by (8), and $\zeta \geq 4$ since $n(4) < 175$. For $k = 1, 2, 3$ the function $U^k/n(\zeta)$ decreases on $[4, \infty)$, because its logarithmic derivative is $k(1 - \frac{1}{8\zeta^2})/U - 1 - \frac{1}{2\zeta} < \frac{k}{U} - 1 < 0$. The same holds for U^4/n , since $U > 4$, and for ζ/n and ζ^2/n . The values at $\zeta = 4$ give, for every $\zeta \geq 4$,

$$r \leq 0.023135, \quad \frac{U^2}{N} \leq 0.093262, \quad \frac{U^3}{N} \leq 0.375960, \quad \frac{U^4}{N} \leq 1.51559, \quad \frac{\zeta(\zeta+1)}{N} \leq 0.114776.$$

We start with the upper bound. Let $W = U/(1-r)$, so that $NW/(N+W) = U$, and let $\delta = W(W+2)/(2N)$. By Lemma 3.5(b) and (9) it suffices to show that $G_U := \psi(W) - \psi(\zeta) + \log E(W) + \log(1-\delta) > 0$. First, $\psi(W) - \psi(\zeta) = (W-U) + (U-\zeta) + \frac{1}{2} \log \frac{U}{\zeta} + \frac{1}{2} \log \frac{W}{U}$, where $W-U = \frac{Ur}{1-r}$, $U-\zeta = \frac{1}{8\zeta}$, $\log \frac{U}{\zeta} = \log(1 + \frac{1}{8\zeta^2}) \geq \frac{1}{8\zeta^2} - \frac{1}{128\zeta^4}$ and $\log \frac{W}{U} = -\log(1-r) \geq r$. Second, Lemma 3.6 gives $E(W) \geq 1-t$ with $t = \frac{1}{8W} + \frac{3}{128W^2} + \frac{45}{512W^3}$. Since $W \geq U \geq \zeta$ and $\frac{1}{8U} = \frac{1}{8\zeta} - \frac{1}{8\zeta(8\zeta^2+1)} \leq \frac{1}{8\zeta} - \frac{1}{64\zeta^3} + \frac{1}{512\zeta^5}$, we get $t \leq \tau := \frac{1}{8\zeta} + \frac{3}{128\zeta^2} + \frac{37}{512\zeta^3} + \frac{1}{512\zeta^5}$ and $\log E(W) \geq \log(1-\tau) \geq -\tau - \frac{\tau^2}{2(1-\tau)}$. Third, $\delta = \frac{Ur}{2(1-r)^2} + \frac{r}{1-r}$ and $\log(1-\delta) \geq -\delta - \frac{\delta^2}{2(1-\delta)}$. Adding the 3 parts gives $G_U \geq B + D$, where

$$B = \frac{5}{128\zeta^2} - \frac{37}{512\zeta^3} - \frac{1}{256\zeta^4} - \frac{1}{512\zeta^5} - \frac{\tau^2}{2(1-\tau)}, \quad D = \frac{r}{2(1-r)} \left(\frac{U(1-2r)}{1-r} - 1 - r \right) - \frac{\delta^2}{2(1-\delta)}.$$

Both $\zeta\tau$ and τ decrease in ζ , so $\zeta^2 B$ increases. At $\zeta = 4$ we have $\zeta\tau = 0.135384$, $\tau = 0.033846$ and $\zeta^2 B = 0.011236$. Hence $B > 0$. In D the bracket is at least $4.03125(1-0.023683) - 1.023135 > 2.912$, so the first term of D is at least $1.456r$. Write $\delta = rm$ with $m = \frac{U}{2(1-r)^2} + \frac{1}{1-r}$. Then $\delta \leq 0.07255$ and $rm^2 = \frac{U^3}{4N(1-r)^4} + \frac{U^2}{N(1-r)^3} + \frac{r}{(1-r)^2} \leq 0.2276$, so $\frac{\delta^2}{2(1-\delta)} = r \cdot \frac{rm^2}{2(1-\delta)} \leq 0.1228r$. Hence $D \geq 1.33r > 0$. So $G_U > 0$ and $z_N < U$.

Now the lower bound. Let $\varepsilon = \varepsilon_N$, $\ell = U - \varepsilon$, $r' = \ell/N$ and $W' = \ell/(1-r')$, so that $NW'/(N+W') = \ell$. By these bounds, $\varepsilon \leq 0.114776 + \frac{1}{256} < 0.1187$, so $\ell \geq \zeta - 0.1187 \geq 3.88$ and $W' \geq \ell$. By Lemma 3.5(a) and (9) it suffices to show that $G_\ell := \psi(W') - \psi(\zeta) + \log E(W') < 0$. First, $\psi(W') - \psi(\zeta) = (\ell - \zeta) + \frac{\ell r'}{1-r'} + \frac{1}{2} \log \frac{\ell}{\zeta} - \frac{1}{2} \log(1-r')$, where $\ell - \zeta = \frac{1}{8\zeta} - \varepsilon$, $\frac{1}{2} \log \frac{\ell}{\zeta} \leq \frac{1}{2}(\frac{\ell}{\zeta} - 1) = \frac{1}{16\zeta^2} - \frac{\varepsilon}{2\zeta}$ and $-\frac{1}{2} \log(1-r') \leq \frac{r'}{2(1-r')}$. The 2 terms in r' add up to $\frac{\ell^2 + \ell/2}{N(1-r')}$. Second, Lemma 3.6 gives

$E(W') \leq 1 - t'$ with $t' = \frac{1}{8W'} + \frac{3}{128W'^2} - \frac{15}{512W'^3} \in (0, 1)$, so $\log E(W') \leq -t'$. Here $\frac{1}{8W'} = \frac{1}{8\ell} - \frac{1}{8N}$. Write $\ell = \zeta(1 + \kappa)$ with $\kappa = \frac{1}{8\zeta^2} - \frac{\varepsilon}{\zeta}$. Then $\frac{1}{\ell} \geq \frac{1-\kappa}{\zeta}$, so $-\frac{1}{8W'} \leq -\frac{1}{8\zeta} + \frac{1}{64\zeta^3} - \frac{\varepsilon}{8\zeta^2} + \frac{1}{8N}$. Next, $x \mapsto x/(1 - x/N)$ increases and $\ell \leq U$, so $W' \leq U/(1 - r)$ and $\frac{1}{W'^2} \geq \frac{(1-r)^2}{\zeta^2(1+\frac{1}{8\zeta^2})^2} \geq \frac{(1-2r)(1-\frac{1}{4\zeta^2})}{\zeta^2}$. Finally $\frac{15}{512W'^3} \leq \frac{15}{512\ell^3}$. Adding, and using $\varepsilon(1+\frac{1}{2\zeta}) = \frac{\zeta^2+\frac{3}{2}\zeta+\frac{1}{2}}{N} + \frac{1}{16\zeta^2} + \frac{1}{32\zeta^3}$, we get $G_\ell \leq \beta + \Delta - \frac{\varepsilon}{8\zeta^2}$ with

$$\beta = -\frac{3}{128\zeta^2}(1-2r)\left(1 - \frac{1}{4\zeta^2}\right) + \frac{15}{512\ell^3} - \frac{1}{64\zeta^3}, \quad \Delta = \frac{1}{N}\left(\frac{\ell^2 + \ell/2}{1-r'} + \frac{1}{8} - \zeta^2 - \frac{3}{2}\zeta - \frac{1}{2}\right).$$

For $\beta < 0$ it suffices that $\zeta^2/\ell^3 \leq \frac{4}{5}(1-2r)(1 - \frac{1}{4\zeta^2})$. The left side is at most $\zeta^2/(\zeta - 0.12)^3 \leq 16/3.88^3 < 0.274$, since this function decreases for $\zeta > 0.12$ (its logarithmic derivative is $\frac{2}{\zeta} - \frac{3}{\zeta-0.12} < 0$ there). The right side is at least $0.8 \cdot 0.9537 \cdot 0.9843 > 0.75$. For $\Delta < 0$, note that $\ell \leq U$ and $r' \leq r$, so

$$\frac{\ell^2 + \ell/2}{1-r'} \leq U^2 + \frac{U}{2} + \frac{U^3/N + U^2/(2N)}{1-r} \leq \zeta^2 + \frac{\zeta}{2} + 0.2667 + 0.4326,$$

using $U^2 + \frac{U}{2} = \zeta^2 + \frac{\zeta}{2} + \frac{1}{4} + \frac{1}{16\zeta} + \frac{1}{64\zeta^2}$. This is less than $\zeta^2 + \frac{3}{2}\zeta + \frac{3}{8}$ because $\zeta \geq 4$. Hence $G_\ell < 0$ and $z_N > \ell$.

The constants were recomputed in 40-digit arithmetic, and as a check outside the proof we evaluated $\zeta^2 B$, D , β and Δ on $\zeta \in [4, 60]$ with step 0.001.

A.3 The certificates

Let $c, d > 0$ be integers, and weight each repeat by d and each change by c . By Theorem 2.1 and Proposition C.1 with $(\alpha, \beta) = (d, c)$,

$$d^{N-1}S_N(c/d) = \sum_j W(N, m_N, j) c^j d^{N-1-j} = n_N(m_N).$$

So $S_N(c/d) < 1$ exactly when the integer $n_N(m_N)$ is less than d^{N-1} .

A.4 Proof of Proposition 3.8

Write $w = w_N$, $z = z_N$, $\Lambda_N = 2\Phi(w)/N$ and $\delta = w(w+2)/(2N)$. At $w = w_N$ the display in the proof of Lemma 3.5 gives $1 \leq \Lambda_N \leq 1/(1-\delta)$ when $\delta < 1$, and (9) gives $\psi(w) + \log E(w) = L + c_0 + \log \Lambda_N$. Since $w - z = wz/N$, $w/z = 1 + w/N$ and $\frac{1}{8z} = \frac{1}{8w} + \frac{1}{8N}$, the remainder $R_N = \psi(z) - L - c_0 - \frac{1}{8z}$ is

$$R_N = \log \Lambda_N - \left(\log E(w) + \frac{1}{8w}\right) - \frac{1}{8N} - \frac{wz}{N} - \frac{1}{2} \log\left(1 + \frac{w}{N}\right). \quad (45)$$

Let $N \geq 175$. By the previous proof, $3.88 \leq z \leq w \leq U/(1-r)$. For the lower bound, note that $\log \Lambda_N \geq 0$ and $-\log E(w) \geq -\log(1 - \frac{1}{8w}) \geq \frac{1}{8w}$ by Lemma 3.6, since $w \geq \frac{5}{4}$. So $R_N \geq -\frac{wz+w/2+1/8}{N} = -\frac{z^2+z/2}{N-z} - \frac{1}{8N}$. This is at least $-\frac{z^2+z}{N}$ as soon as $N \geq \frac{2z^2(z+1)}{z-1/4}$. The right side increases in z and $z \leq U$, so it suffices that $U^2/N \leq \frac{U-1/4}{2(U+1)}$. The left side is at most 0.0933 and the right side at least 0.3757.

For the upper bound let $t = \frac{1}{8w} + \frac{3}{128w^2} + \frac{45}{512w^3} \leq \frac{1.09504}{8w}$, which holds as $w \geq 3.88$. Lemma 3.6 and $-\log(1-t) \leq t + \frac{t^2}{2(1-t)}$ give

$$-\log E(w) - \frac{1}{8w} \leq \frac{3}{128w^2} + \frac{45}{512w^3} + \frac{t^2}{2(1-t)} \leq \frac{0.0559}{w^2} \leq \frac{0.0559}{z^2}.$$

Using $\log \Lambda_N \leq \delta + \frac{\delta^2}{2(1-\delta)}$, $\log(1+x) \geq x - \frac{x^2}{2}$ and $wz \geq w^2 - w^3/N$, the rest of (45) is at most $-\frac{(w-1/2)^2}{2N} + \frac{w^3+w^2/4}{N^2} + \frac{\delta^2}{2(1-\delta)}$. By the bounds on r and U^k/N above, with $w \leq U/(1-r)$, $\frac{w^3+w^2/4}{N} \leq 0.4278$ and $N \frac{\delta^2}{2(1-\delta)} = \frac{(w^4+4w^3+4w^2)/(4N)}{2(1-\delta)} \leq 0.4945$, while $\frac{(w-1/2)^2}{2} \geq 5.71$. So the rest is negative and $R_N \leq 0.0559/z^2 < \frac{1}{16z^2}$.

For the L -form, $z < U = L + c_0 - \frac{1}{2} \log \zeta + \frac{1}{8\zeta} < L$. Also $z > \zeta - 0.1187 \geq \nu_N - 0.1187 \geq L/2$, because $L - \log L \geq 1.18$ for $L \geq \log 175$. Then $\frac{1}{16z^2} \leq \frac{1}{4L^2}$ and $\frac{z^2+z}{N} \leq \frac{L^2+L}{N}$.

Now let $N \leq 174$. The function $R(z) = \psi(z) - L - c_0 - \frac{1}{8z}$ increases in z , while $\frac{1}{16z^2}$ and $-\frac{z^2+z}{N}$ decrease. So it suffices to check $R(z(c'/d)) \leq \frac{1}{16z(c'/d)^2}$ and $R(z(c/d)) \geq -\frac{z(c/d)^2+z(c/d)}{N}$ with the certificates, together with $z(c/d) \geq L/2$ and, for $N \geq 3$, $z(c'/d) \leq L$. All hold, and the smallest gaps are 0.0374 (upper) and 0.0813 (lower), both at $N = 174$. For $N = 2$ we have $S_2(u) = 2u$, so $u_2 = \frac{1}{2}$ and $z_2 = \frac{2}{3}$, and $\frac{2}{3} < \log 2$ is equivalent to $e^2 < 8$, which holds since $e^2 < 2.72^2 = 7.3984$.

B The uniform local law

We keep the notation of Section 6, with $\tilde{I}_j(x) = e^{-x} I_j(x)$.

Lemma B.1 (tilting). *For $s > 0$ let $Y_1 \sim \text{Bin}(n_1, r_1)$ and $Y_2 \sim \text{Bin}(n_2, r_2)$ be independent under $\mathbb{P}_s$. Then $\mathbb{P}(D = j) = \Lambda(s)s^{-j}\mathbb{P}_s(D = j)$ for every $j \in \mathbb{Z}$.*

Proof. Note that $\mathbb{P}(Y_1 = r)s^r = \binom{n_1}{r}(qs)^r p^{n_1-r} = (p + qs)^{n_1} \mathbb{P}_s(Y_1 = r)$, and likewise $\mathbb{P}(Y_2 = r)s^{-r} = (p + q/s)^{n_2} \mathbb{P}_s(Y_2 = r)$. Hence $\mathbb{P}(D = j) = s^{-j} \sum_r \mathbb{P}(Y_1 = r)s^r \mathbb{P}(Y_2 = r - j)s^{j-r} = \Lambda(s)s^{-j}\mathbb{P}_s(D = j)$. $\square$

Lemma B.2 (saddle). *If $n_1, n_2 \geq 1$, then $\mathbb{E}_s D = n_1 r_1 - n_2 r_2$ increases strictly from $-n_2$ to n_1 as s goes from 0 to ∞ . So it vanishes at a unique $s > 0$, the positive root of $n_1 p s^2 + (n_1 - n_2) q s - n_2 p = 0$, and $s \geq 1$ if and only if $n_1 \leq n_2$.*

Proof. Note that r_1 increases from 0 to 1 and r_2 decreases from 1 to 0. Clearing denominators in $n_1 q s / (p + q s) = n_2 q / (p s + q)$ gives the quadratic, and at $s = 1$ the difference is $(n_1 - n_2)q$. $\square$

From now on $2 \leq k \leq N - 2$ and s is this root.

Lemma B.3 (Fourier form). *Let $\chi(t) = \mathbb{E}_s e^{itD} = (1 - r_1 + r_1 e^{it})^{n_1} (1 - r_2 + r_2 e^{-it})^{n_2}$, $g(t) = e^{x(\cos t - 1)}$ and $\omega(t) = 2 + s e^{it} + s^{-1} e^{-it}$. Then*

$$P_N(k) = \frac{q}{2} \Lambda(s) \frac{1}{2\pi} \int_{-\pi}^{\pi} \chi \omega dt, \quad U_N(k) = \frac{q}{2} \Lambda(s) \frac{1}{2\pi} \int_{-\pi}^{\pi} g \omega dt.$$

Proof. By Theorem 6.1 and Lemma B.1, $P_N(k) = \frac{q}{2} \Lambda(s) [2\mathbb{P}_s(D = 0) + s^{-1}\mathbb{P}_s(D = 1) + s\mathbb{P}_s(D = -1)]$, and $\mathbb{P}_s(D = j) = \frac{1}{2\pi} \int \chi e^{-ijt} dt$. Since g is even and $I_j(x) = \frac{1}{\pi} \int_0^\pi e^{x \cos t} \cos(jt) dt$ [28, Eq. 10.32.3], we have $\frac{1}{2\pi} \int g dt = \tilde{I}_0(x)$ and $\frac{1}{2\pi} \int g e^{\pm it} dt = \tilde{I}_1(x)$. $\square$

Lemma B.4 (modulus). *We have $|\chi| \leq g$ and $g - |\chi| \leq g \min(1, 4\kappa^2 V)$.*

Proof. Note that $|1 - r + re^{\pm it}|^2 = 1 - 2r(1 - r)\kappa$. Put $f(z) = \sqrt{1 - z}e^{z/2}$ and $z_i = 2v_i\kappa \in [0, 1]$ ($v_i \leq \frac{1}{4}$, $\kappa \leq 2$). As $x = n_1v_1 + n_2v_2$, we get $|\chi|/g = f(z_1)^{n_1}f(z_2)^{n_2}$. Now $f(0) = 1$ and $f'(z) = -\frac{z}{2}e^{z/2}(1 - z)^{-1/2} \leq 0$, so $0 \leq f \leq 1$. Also $f(z) \geq 1 - z^2$, since for $z < 1$ this says $e^z \geq (1 - z)(1 + z)^2 = 1 + z - z^2 - z^3$, which follows from $e^z \geq 1 + z$. So Lemma 3.2(a), with the $n_1 + n_2$ numbers $1 - f(z_i) \leq z_i^2$, gives $1 - |\chi|/g \leq n_1z_1^2 + n_2z_2^2 = 4\kappa^2V$. $\square$

Lemma B.5 (phase). *For $|t| < \pi$ write $\chi(t) = |\chi(t)|e^{i\Theta(t)}$ with Θ continuous and $\Theta(0) = 0$. Then $|\Theta(t)| \leq c_\varphi M|t|^3$ with $c_\varphi = 128/(3\pi^3) = 1.3761\dots$*

Proof. For $0 \leq r \leq 1$ and $|t| < \pi$, $w_r(t) = 1 - r + re^{it}$ lies on the segment from 1 to e^{it} , so $w_r \neq 0$ and $\arg w_r(t)$ lies between 0 and t . Writing $w_r(t) = e^{it/2}[\cos \frac{t}{2} + i(2r - 1)\sin \frac{t}{2}]$ gives $\arg w_r(t) = \frac{t}{2} - \arctan((1 - 2r)\tan \frac{t}{2})$. As the factor for $-Y_2$ is $\overline{w_{r_2}}$, $\Theta = n_1 \arg w_{r_1} - n_2 \arg w_{r_2}$. Subtracting $(n_1r_1 - n_2r_2)\sin t = 0$ gives $\Theta = n_1\varphi(r_1, t) - n_2\varphi(r_2, t)$ with $\varphi(r, t) = \arg w_r(t) - r \sin t$. Both are odd in t , so let $0 < t < \pi$. We show that $0 \leq \varphi(r, t) \leq c_\varphi r^2 t^3$.

Put $\tau = \tan \frac{t}{2}$, $b_0 = 1 - 2r$ and $\gamma(b) = \arctan(b\tau) - b\tau/(1 + \tau^2)$. Since $\frac{t}{2} = \arctan \tau$ and $\sin t = 2\tau/(1 + \tau^2)$, we get $\varphi = \gamma(1) - \gamma(b_0)$. Also γ is odd and $\gamma'(b) = \tau^3(1 - b^2)/[(1 + \tau^2)(1 + b^2\tau^2)] \geq 0$ on $[-1, 1]$. Hence

$$\varphi = \frac{\tau^3}{1 + \tau^2} \int_{b_0}^1 \frac{1 - b^2}{1 + b^2\tau^2} db \geq 0, \quad \int_{b_0}^1 (1 - b^2) db = 4r^2 - \frac{8}{3}r^3 \leq 4r^2.$$

There are 4 cases. (a) If $t \leq \frac{\pi}{2}$, then $\tau \leq 2t/\pi \leq 1$ by convexity of $\tan$ on $[0, \frac{\pi}{4}]$, so $\varphi \leq 4r^2\tau^3 \leq \frac{32}{\pi^3}r^2t^3$. (b) If $t > \frac{\pi}{2}$ and $r \leq \frac{1}{4}$, then $b_0 \geq \frac{1}{2}$, so $1 + b^2\tau^2 \geq 1 + \tau^2/4$ and $\varphi \leq 16r^2\tau^3/[(1 + \tau^2)(4 + \tau^2)]$. As $(1 + \tau^2)(4 + \tau^2) \geq \max(9\tau^2, \tau^4)$, this is at most $16r^2 \min(\tau/9, 1/\tau) \leq \frac{16}{3}r^2 \leq c_\varphi r^2 t^3$, as $t^3 > \pi^3/8$. (c) If $t > \frac{\pi}{2}$ and $\frac{1}{4} < r \leq \frac{1}{2}$, then $\gamma(b_0) \geq \gamma(0) = 0$, so $\varphi \leq \gamma(1) = \frac{1}{2}(t - \sin t) \leq \frac{t^3}{12} < \frac{4}{3}r^2 t^3$. (d) If $r > \frac{1}{2}$, then $\varphi = \gamma(1) + \gamma(|b_0|) \leq 2\gamma(1) = t - \sin t \leq \frac{t^3}{6} < \frac{2}{3}r^2 t^3$. As $\frac{32}{\pi^3}, \frac{4}{3}, \frac{2}{3} < c_\varphi$, the claim follows. Finally $\varphi(r_1, t)$ and $\varphi(r_2, t)$ have the same sign, so $|\Theta| \leq \max_i n_i \varphi(r_i, t) \leq c_\varphi |t|^3 \max_i n_i r_i^2 = c_\varphi M|t|^3$. $\square$

Lemma B.6 (moments). *For every $x > 0$, with $\mathcal{R} = \mathcal{R}(x)$,*

$$(i) \quad \langle \kappa^2 \rangle = 2 - 2\mathcal{R} - \mathcal{R}/x \text{ and } \langle \kappa^3 \rangle = 4 - 4\mathcal{R} - 3\mathcal{R}/x - 2\mathcal{R}/x^2 + 1/x,$$

$$(ii) \quad 1/\mathcal{R}(x) \leq 1 + 2/x,$$

$$(iii) \quad \langle \kappa^2 \rangle \leq \min(\frac{3}{2}, 2/x^2) \text{ and } \langle \kappa^3 \rangle \leq \min(\frac{5}{2}, 8/x^3).$$

Proof. (i) Note that $\kappa^2 = \frac{3}{2} - 2\cos t + \frac{1}{2}\cos 2t$, $\kappa^3 = \frac{5}{2} - \frac{15}{4}\cos t + \frac{3}{2}\cos 2t - \frac{1}{4}\cos 3t$ and $\langle \cos jt \rangle = I_j(x)/I_0(x)$ [28, Eq. 10.32.3]. The recurrence $I_{\nu-1} - I_{\nu+1} = (2\nu/x)I_\nu$ [28, Eq. 10.29.1] gives $I_2/I_0 = 1 - 2\mathcal{R}/x$ and $I_3/I_0 = \mathcal{R} - 4/x + 8\mathcal{R}/x^2$.

(ii), (iii) By [28, Eqs. 10.29.2, 10.29.3], $I'_0 = I_1$ and $I'_1 = I_0 - I_1/x$, so $\mathcal{R}' = 1 - \mathcal{R}/x - \mathcal{R}^2$. For a C^1 function h put $d_h = 1 - h/x - h^2 - h'$. If $d_h > 0$ on $[x_*, \infty)$ and $\mathcal{R}(x_*) > h(x_*)$, then $\mathcal{R} > h$ on $[x_*, \infty)$. Indeed, at a first x_1 with $\mathcal{R}(x_1) = h(x_1)$ we would have $(\mathcal{R} - h)'(x_1) \leq 0$, but it is $d_h(x_1) > 0$. Each d_h below follows by clearing denominators.

First, $h_0 = x/(x + 2)$ has $d_{h_0} = 3x/(x + 2)^2 > 0$, and $\mathcal{R} - h_0 = \frac{x^2}{4} + O(x^3)$ as $x \rightarrow 0$ since $\mathcal{R} = \frac{x}{2} - \frac{x^3}{16} + O(x^5)$. So $\mathcal{R} > h_0$ for all $x > 0$, which is (ii). It also gives $\langle \kappa^2 \rangle \leq 2 - (2 + \frac{1}{x})h_0 = \frac{3}{x+2} \leq \frac{3}{2}$.

Second, by (i), $\langle \kappa^2 \rangle < 2/x^2$ means $\mathcal{R} > h_2 = (2x^2 - 2)/(x(2x + 1))$. Here $d_{h_2} = (5x^2 - 4x - 4)/(x^2(2x + 1)^2) > 0$ for $x > 1.3798$, and at $x_* = 1.38$ we have $\mathcal{R} > h_0 = 0.40828 > h_2 = 0.34860$. For $x < 1.38$, $\frac{3}{x+2} \leq \frac{2}{x^2}$ since $3x^2 - 2x - 4 \leq 0$ for $x \leq 1.535$.

Third, $\langle \kappa^3 \rangle < 8/x^3$ means $\mathcal{R} > h_3 = (4x^3 + x^2 - 8)/(x(4x^2 + 3x + 2))$. Here $d_{h_3} = (49x^3 - 48x^2 - 24x - 64)/(x^2(4x^2 + 3x + 2)^2) > 0$ for $x \geq 1.72$, and at $x_* = 1.72$ we have $\mathcal{R} > h_2 = 0.51289 > h_3 = 0.46870$. For $x < 1.72$, $\langle \kappa^3 \rangle \leq 2\langle \kappa^2 \rangle \leq 4/x^2 \leq 8/x^3$.

Finally, the density is proportional to $e^{-x\kappa}$, so $\frac{d}{dx}\langle \kappa^j \rangle = \langle \kappa^j \rangle \langle \kappa \rangle - \langle \kappa^{j+1} \rangle \leq 0$ by Chebyshev's sum inequality for the increasing functions κ^j and κ . As $x \rightarrow 0$ the averages tend to $\frac{1}{2\pi} \int (1 - \cos t)^2 dt = \frac{3}{2}$ and $\frac{1}{2\pi} \int (1 - \cos t)^3 dt = \frac{5}{2}$. $\square$

Proof of Theorem 6.2. By Lemma B.3, $\theta = \Delta/(2\tilde{I}_0 + (s + s^{-1})\tilde{I}_1)$ with $\Delta = \frac{1}{2\pi} \int (\chi - g)\omega dt$. Both integrals there are real, so Δ integrates

$$\operatorname{Re}[(\chi - g)\omega] = (|\chi| \cos \Theta - g)(2 + (s + s^{-1}) \cos t) - |\chi| \sin \Theta (s - s^{-1}) \sin t.$$

Now $g - |\chi| \cos \Theta = (g - |\chi|) + |\chi|(1 - \cos \Theta)$ lies between 0 and $g\mu$ with $\mu = \min(1, 4\kappa^2 V) + \frac{1}{2}\Theta^2$, by Lemma B.4 and $1 - \cos \Theta \leq \frac{1}{2}\Theta^2$. Also $|\chi \sin \Theta| \leq g|\Theta|$ and $|s - s^{-1}| \leq s + s^{-1}$. Since $\frac{1}{2\pi} \int gh dt = \tilde{I}_0 \langle h \rangle$,

$$|\Delta| \leq 2\tilde{I}_0 \langle \mu \rangle + (s + s^{-1})\tilde{I}_0 [\langle \mu |\cos t| \rangle + \langle |\Theta| |\sin t| \rangle].$$

Since $|\cos t| \leq 1$ and $\tilde{I}_0/\tilde{I}_1 = 1/\mathcal{R} \geq 1$, the mediant inequality gives

$$|\theta| \leq \mathcal{E}_0 := \frac{\langle \min(1, 4\kappa^2 V) \rangle + \frac{1}{2}\langle \Theta^2 \rangle + \langle |\Theta| |\sin t| \rangle}{\mathcal{R}(x)}.$$

For $\mathcal{E}_0 \leq \mathcal{E}$ we use $\min(1, 4\kappa^2 V) \leq 4\kappa^2 V$, Lemma B.5, $|t| \leq \pi |\sin \frac{t}{2}| = \pi(\kappa/2)^{1/2}$ and $|\sin t| \leq 2(\kappa/2)^{1/2}$. Since $c_\varphi \pi^3 = \frac{128}{3}$, we get $\frac{1}{2}\Theta^2 \leq \frac{1}{2}c_\varphi^2 \pi^6 M^2 (\kappa/2)^3 = \frac{1024}{9} M^2 \kappa^3$ and $|\Theta| |\sin t| \leq 2c_\varphi \pi^3 M (\kappa/2)^2 = \frac{64}{3} M \kappa^2$. $\square$

Proof of Corollary 6.3. Note that $r_1 = r_2 = q$ at $s = 1$, $r_2 \leq q$ if $s \geq 1$, and $r_1 \leq q$ if $s \leq 1$. So $\min(r_1, r_2) \leq q \leq \frac{1}{2}$ and $x = m(1 - r_1) + m(1 - r_2) \geq m/2$. Also $n_i v_i \leq x$, so $v_i \leq \min(x/n_{\min}, r_{\max})$ and $V \leq x \max_i v_i \leq \min(x^2/n_{\min}, x r_{\max})$. Since $r_i = m/n_i$, $M = m r_{\max} = m^2/n_{\min} \leq \min(4x^2/n_{\min}, 2x r_{\max})$. We put these and Lemma B.6 into $\mathcal{E}$.

For the $n_{\min}$ form, the first term of $\mathcal{E}$ is at most $\frac{4+256/3}{n_{\min}}(x^2 + 2x) \min(\frac{3}{2}, \frac{2}{x^2}) \leq \frac{(4+256/3)(2+2\sqrt{3})}{n_{\min}} \leq \frac{488.13}{n_{\min}}$, with the supremum at $x^2 = \frac{4}{3}$ (the product is 488.126...). If $x \leq 1$, the M^2 term is at most $\frac{3}{x} \cdot \frac{1024}{9} \cdot \frac{16x^4}{n_{\min}^2} \cdot \frac{5}{2} \leq \frac{13654}{n_{\min}^2}$. If $x \geq 1$, then $M^2 \leq \min(16x^4/n_{\min}^2, 4x^2)$, and the term is at most

$$3 \cdot \frac{1024}{9} \min\left(\frac{16x^4}{n_{\min}^2}, 4x^2\right) \frac{8}{x^3} = \frac{1024}{3} \min\left(\frac{128x}{n_{\min}^2}, \frac{32}{x}\right) \leq \frac{1024}{3} \cdot \frac{64}{n_{\min}} = \frac{65536/3}{n_{\min}} \leq \frac{21845.34}{n_{\min}},$$

since a minimum is at most the geometric mean. So both cases give at most $21845.34/n_{\min}$, and $488.13 + 21845.34 = 22333.47 \leq 22334$.

For the $r_{\max}$ form, the first term is at most $(4 + \frac{128}{3})r_{\max} \sup_x (x + 2) \min(\frac{3}{2}, \frac{2}{x^2}) = (4 + \frac{128}{3})(3 + \sqrt{3})r_{\max} \leq 220.83 r_{\max}$ (the product is 220.829..., with the supremum at $x^2 = \frac{4}{3}$). The M^2 term is at most $\frac{4096}{9} r_{\max}^2 \sup_x (x^2 + 2x) \min(\frac{5}{2}, \frac{8}{x^3})$, where the supremum is 12.7968... at $x^3 = 3.2$. So the term is at most $\frac{4096}{9} \cdot 12.797 r_{\max}^2 \leq 5825 r_{\max}^2$.

Finally, by symmetry let $n_1 \leq n_2$, and put $u = q/p \leq 2q$. Then $s \geq 1$, so $r_{\max} = r_1 \leq qs/p = us$. By Lemma B.2,

$$s = \frac{(n_2 - n_1)u}{2n_1} + \sqrt{\frac{(n_2 - n_1)^2 u^2}{4n_1^2} + \frac{n_2}{n_1}} \leq \frac{(n_2 - n_1)u}{n_1} + \sqrt{\frac{n_2}{n_1}}.$$

With $n_1 \geq 1$ and $n_2 \leq N$ this gives $r_{\max} \leq 2q\sqrt{N} + 4q^2N = \delta_T$. $\square$

C The numerator tables

Proposition C.1. *For $p = \alpha/(\alpha + \beta)$ and $N \geq 1$, the numerators of Remark 2.3 are $n_N(k) = \sum_w \alpha^{\text{rep}(w)} \beta^{\text{chg}(w)}$, summed over the words with k letters R and $N - k$ letters L. So they are integers, and*

$$\sum_{a,b \geq 0} n_{a+b}(a) x^a y^b = \frac{1 - (\alpha - 1)(x + y) + (\alpha - \beta)(\alpha + \beta - 2)xy}{1 - \alpha(x + y) + (\alpha^2 - \beta^2)xy}.$$

Proof. Multiply (1) by $2(\alpha + \beta)^{N-1}$. Let H_R and H_L be the weighted sums over the words ending in R and in L, with x marking R and y marking L. Then $H_R = x(1 + \alpha H_R + \beta H_L)$ and $H_L = y(1 + \alpha H_L + \beta H_R)$, and solving as in the proof of Proposition 2.2 gives $1 + H_R + H_L$. $\square$

For $(\alpha, \beta) = (2, 1)$ this is $(1 - x)(1 - y)/(1 - 2x - 2y + 3xy)$, and for $(1, 2)$ it is $(1 - xy)/(1 - x - y - 3xy)$, the generating functions recorded in A035002 and A348595 [22, 23]. Figure 6 shows both bijections.

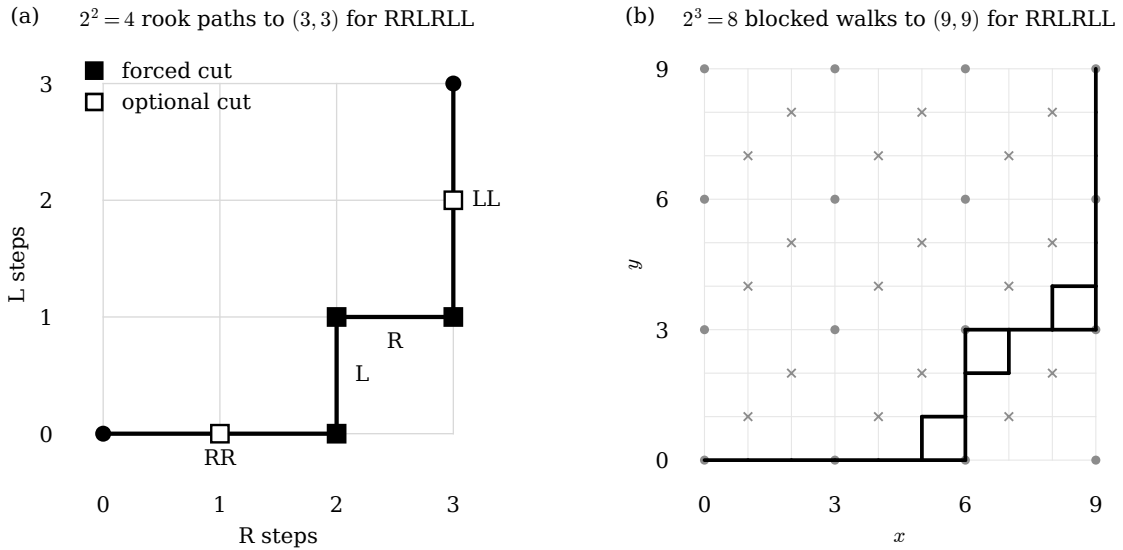


Figure 6: The 2 bijections for the word RRLRLL. (a) The rook path. A cut is forced at each of the 3 direction changes (filled) and optional between equal letters (hollow), so the word comes from $2^2 = 4$ rook paths to $(3, 3)$. (b) The union of the $2^3 = 8$ blocked walks to $(9, 9)$ that give the word, with 1 detour square for each direction change. Blocked points are marked $\times$ and corners by dots.

Theorem C.2 ($p = \frac{2}{3}$ and rook paths). *Let $r(a, b)$ count the rook paths from $(0, 0)$ to (a, b) with every move to the right or up. For $p = \frac{2}{3}$, $n_N(k) = r(k, N - k)$ for all $0 \leq k \leq N$. Since the array of A035002 starts at $(1, 1)$, Kagey’s $p = \frac{2}{3}$ table read by rows is A035002 read by antidiagonals.*

Proof. By definition each entry of A035002 is the sum of the entries to its left and above it, which is the recursion for r obtained by removing the last move. A rook path to (a, b) with $a + b = N \geq 1$ gives a word w with a letters R and b letters L, one letter per unit of length, and a set of cuts $i < N$ where one move ends and the next begins. A cut is forced where $w_i \neq w_{i+1}$ and optional where $w_i = w_{i+1}$, and each choice of optional cuts gives exactly one rook path. Hence, $r(a, b) = \sum_w 2^{\text{rep}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (2, 1)$ by Proposition C.1. $\square$

Theorem C.3 ($p = \frac{1}{3}$ and Mathar’s blocked walks). *Call (x, y) blocked if $x \equiv y \equiv 1$ or $x \equiv y \equiv 2 \pmod{3}$, and let $M(a, b)$ count the walks from $(0, 0)$ to $(3a, 3b)$ with unit steps right and up that avoid blocked points. For $p = \frac{1}{3}$, $n_N(k) = M(k, N - k)$. The array M is symmetric.*

Proof. Work modulo 3 and call a point with residue $(0, 0)$ a *corner*. Since $(1, 1)$ and $(2, 2)$ are blocked, a right step from a corner is followed by another, to $(2, 0)$, which we call state A, and an up step leads on to $(0, 2)$, state B. From A the walk steps right to a corner, or it goes up, right and up to B, and B behaves symmetrically. Leaving A crosses one line $x \equiv 0$ and no line $y \equiv 0$, leaving B does the reverse, and leaving a corner crosses neither. A walk to $(3a, 3b)$ crosses a lines $x \equiv 0$ and b lines $y \equiv 0$, so writing R for each visit to A and L for each visit to B gives a word w with a letters R and b letters L. Between w_i and w_{i+1} the walk passes through a corner if $w_i = w_{i+1}$, in 1 way, and through a corner or directly if $w_i \neq w_{i+1}$, in 2 ways. The start and the end are forced. Hence $M(a, b) = \sum_w 2^{\text{chg}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (1, 2)$ by Proposition C.1. Swapping the letters R and L keeps $\text{chg}(w)$, so $M(a, b) = M(b, a)$. $\square$

In Mathar’s note [21] the “squares” of A348595 are lattice points. A direct count of the walks gives its published terms (computer-checked, Section 7). So A348595 lists the lower triangle of M , which is how that entry is “related” to the $p = \frac{1}{3}$ table.

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